US012406965B2

(12) **United States Patent**
**Chen et al.**

(10) **Patent No.: US 12,406,965 B2**
(45) **Date of Patent: *Sep. 2, 2025**

(54) **PACKAGE**

(71) Applicant: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

(72) Inventors: **Ming-Fa Chen**, Taichung (TW); **Chao-Wen Shih**, Hsinchu County (TW); **Hsien-Wei Chen**, Hsinchu (TW); **Sung-Feng Yeh**, Taipei (TW); **Tzuan-Horng Liu**, Taoyuan (TW)

(73) Assignee: **Taiwan Semiconductor Manufacturing Company, Ltd.**, Hsinchu (TW)

( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/751,359**

(22) Filed: **Jun. 24, 2024**

(65) **Prior Publication Data**

US 2024/0347512 A1 Oct. 17, 2024

**Related U.S. Application Data**

(60) Continuation of application No. 17/344,928, filed on Jun. 10, 2021, now Pat. No. 12,046,579, which is a (Continued)

(51) **Int. Cl.**
***H01L 25/065*** (2023.01)
***H01L 21/304*** (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ..... ***H01L 25/0657*** (2013.01); ***H01L 21/304*** (2013.01); ***H01L 21/56*** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC ... H01L 25/0657; H01L 21/304; H01L 21/56; H01L 23/3157; H01L 23/5384; H01L 23/544; H01L 24/05; H01L 21/6835; H01L 2224/0231; H01L 2224/02372; H01L 2224/80203; H01L 2224/8084; H01L 2225/06541; H01L 24/08;
(Continued)

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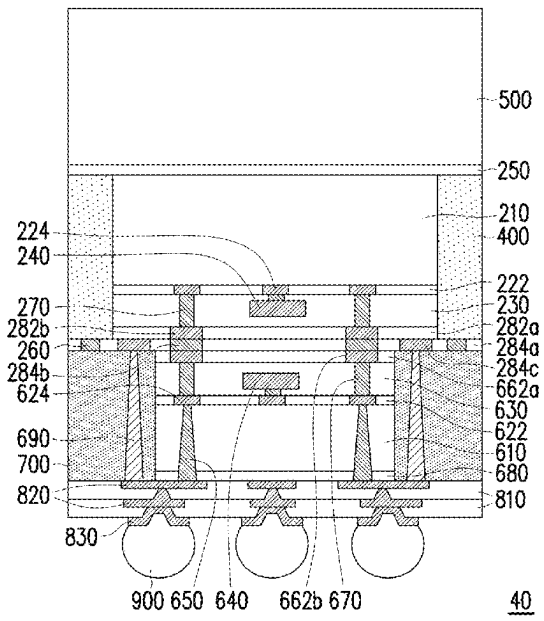
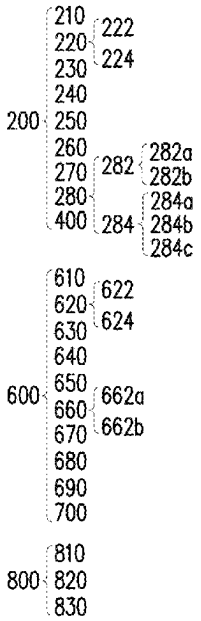
*Primary Examiner* — Jonathan Han
(74) *Attorney, Agent, or Firm* — JCIPRNET

(57) **ABSTRACT**

A package includes a carrier substrate, a first die, and a second die. The first die and the second die are stacked on the carrier substrate in sequential order. The first die includes a first bonding layer, a second bonding layer, and an alignment mark embedded in the first bonding layer. The second die includes a third bonding layer. A surface of the first bonding layer form a rear surface of the first die and a surface of the second bonding layer form an active surface of the first die. The rear surface of the first die is in physical contact with the carrier substrate. The active surface of the first die is in physical contact with the third bonding layer of the second die.

**20 Claims, 25 Drawing Sheets**



**Related U.S. Application Data**

division of application No. 16/572,622, filed on Sep. 17, 2019, now Pat. No. 11,063,022.

(51) **Int. Cl.**

| | |
|---|---|
| ***H01L 21/56*** | (2006.01) |
| ***H01L 21/683*** | (2006.01) |
| ***H01L 23/00*** | (2006.01) |
| ***H01L 23/31*** | (2006.01) |
| ***H01L 23/538*** | (2006.01) |
| ***H01L 23/544*** | (2006.01) |
| ***H10D 86/01*** | (2025.01) |

(52) **U.S. Cl.**

CPC ..... ***H01L 23/3157*** (2013.01); ***H01L 23/5384*** (2013.01); ***H01L 23/544*** (2013.01); ***H01L 24/05*** (2013.01); *H01L 21/6835* (2013.01); *H01L 2224/0231* (2013.01); *H01L 2224/02372* (2013.01); *H01L 2224/80203* (2013.01); *H01L 2224/8084* (2013.01); *H01L 2225/06541* (2013.01); *H10D 86/0214* (2025.01)

(58) **Field of Classification Search**

CPC . H01L 2224/05624; H01L 2224/05647; H01L 2224/05655; H01L 2224/05666; H01L 2224/05684; H01L 2224/08057; H01L 2224/08145; H01L 2224/19; H01L 2224/215; H01L 2224/29186; H01L 2224/73251; H01L 2224/80006; H01L 2224/8001; H01L 2224/80013; H01L 2224/80048; H01L 2224/80097; H01L 2224/8013; H01L 2224/80379; H01L 2224/80906; H01L 2224/80948; H01L 2224/83405; H01L 2224/83409; H01L 2224/83417; H01L 2224/83486; H01L 2224/83493; H01L 2224/9222; H01L 2224/95; H01L 24/80; H01L 24/92; H01L 24/95; H01L 24/96; H01L 2224/80007; H01L 2224/80132; H01L 2224/83896; H01L 2224/92242; H01L 2221/68345; H01L 2221/68359; H01L 2224/18; H01L 2224/20; H01L 2224/96; H01L 2225/06524; H01L 2225/06548; H01L 25/50; H01L 2223/54426; H10D 86/0214

See application file for complete search history.

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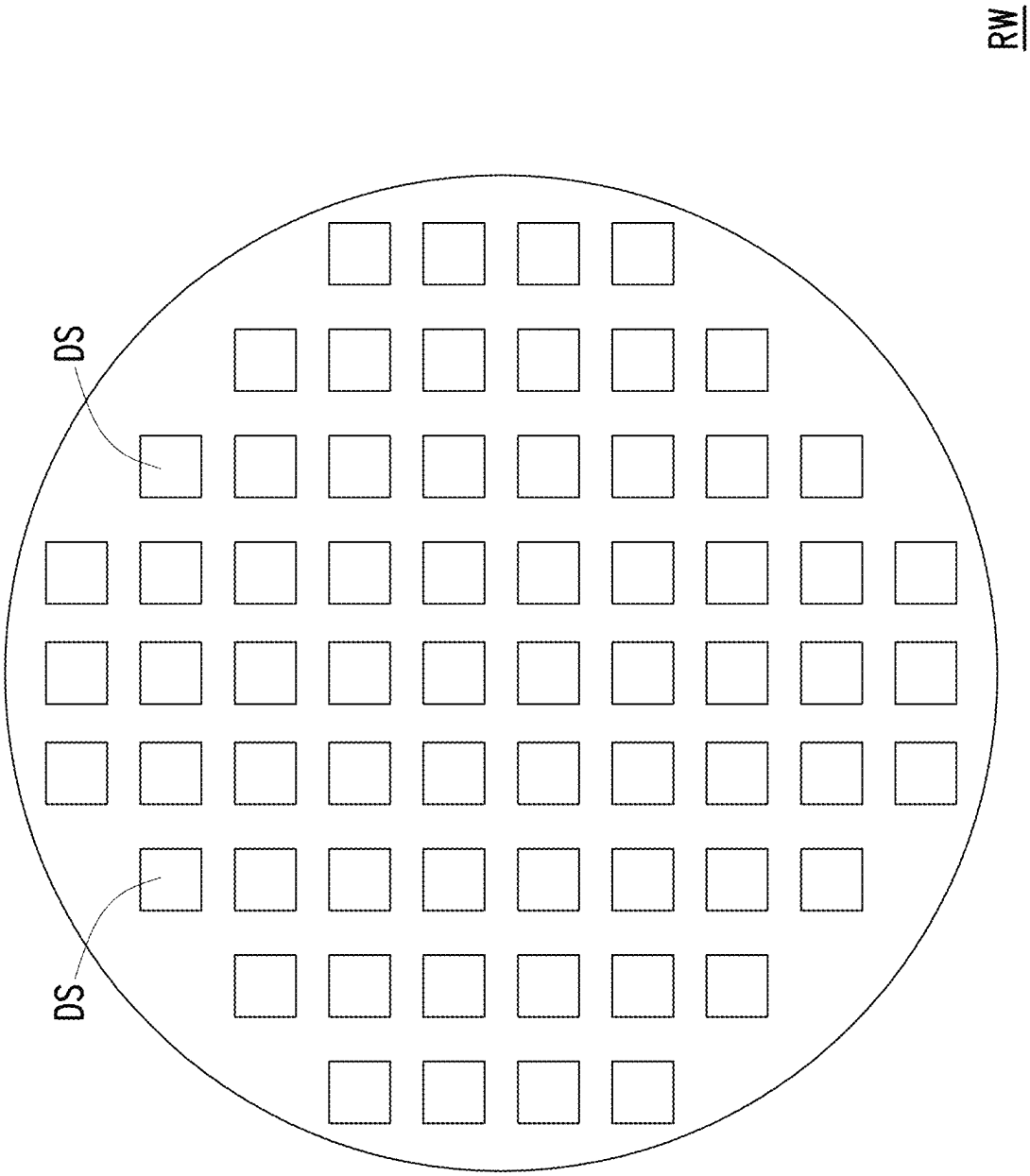
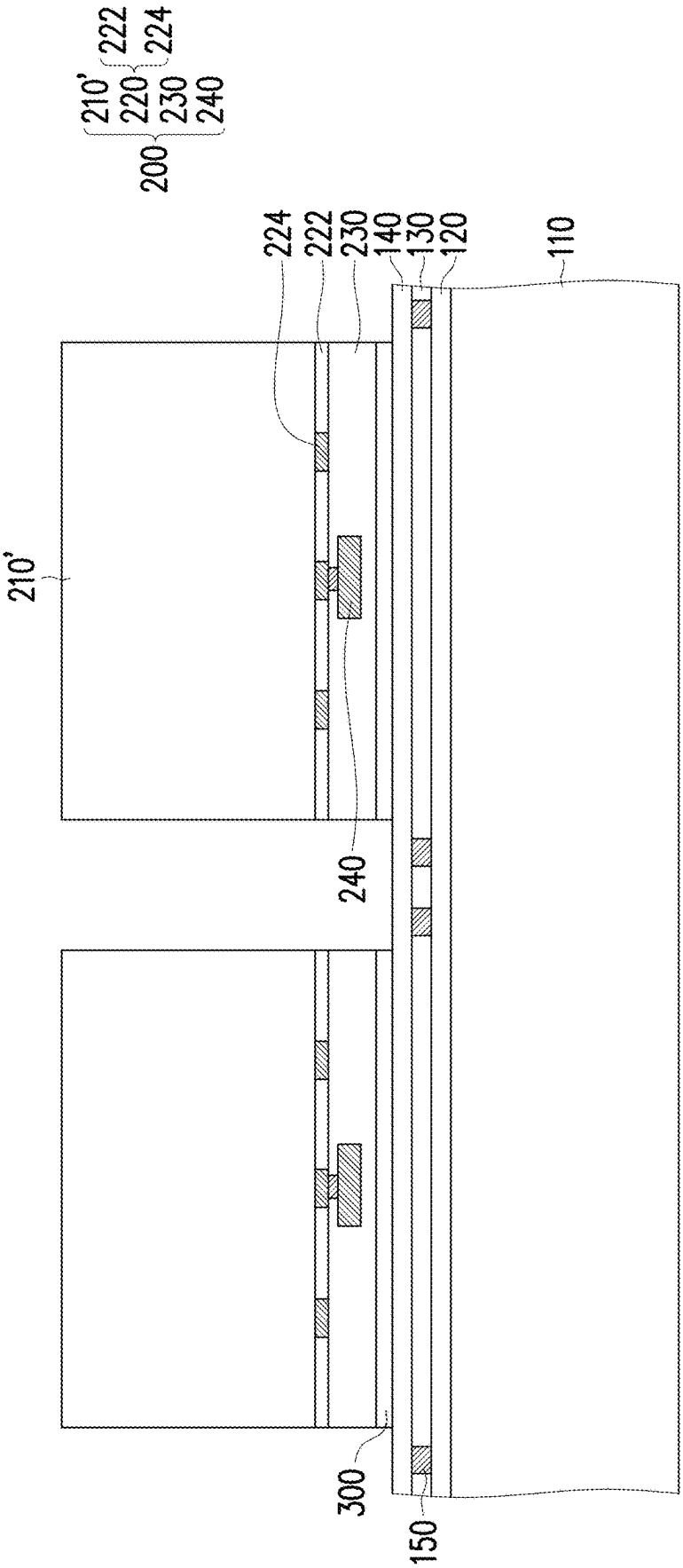



FIG. 1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222
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210'
224
222
230
140
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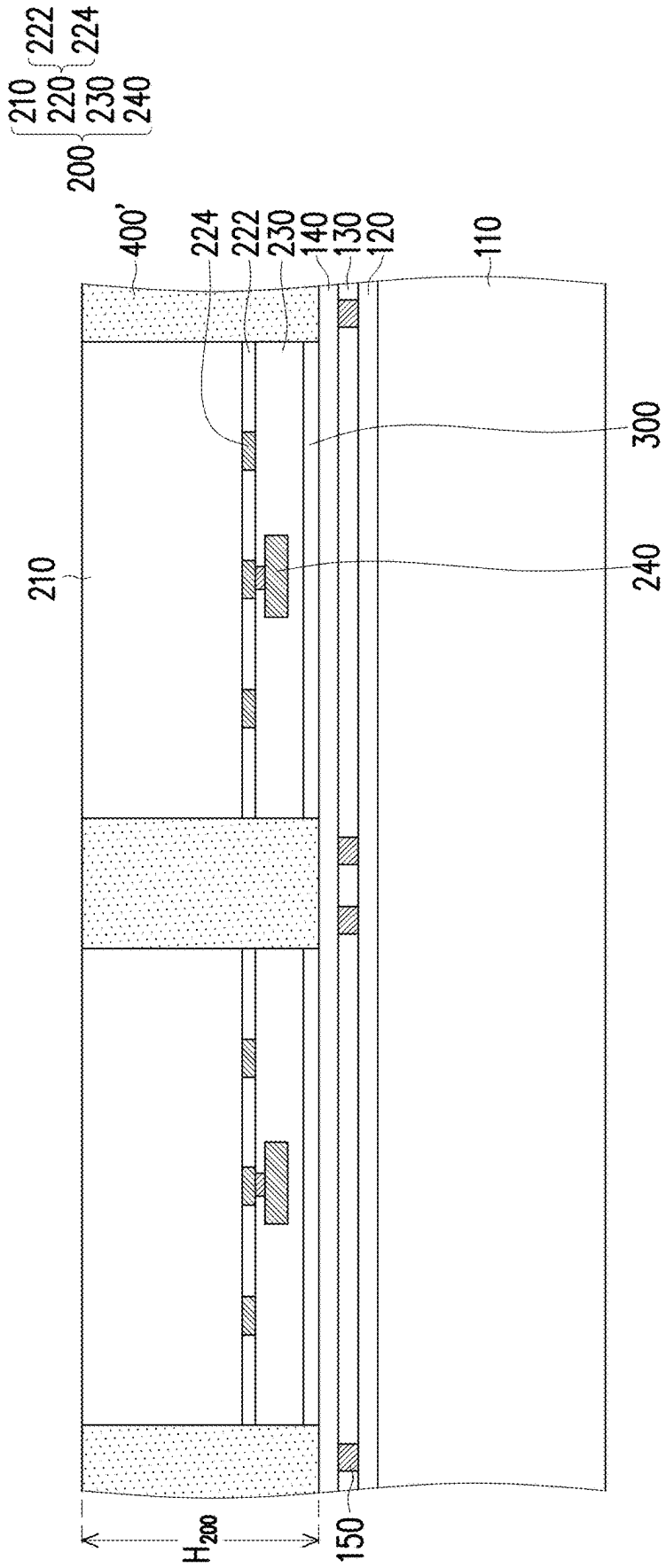
120
110
240
300
150

FIG. 2A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210
220
222
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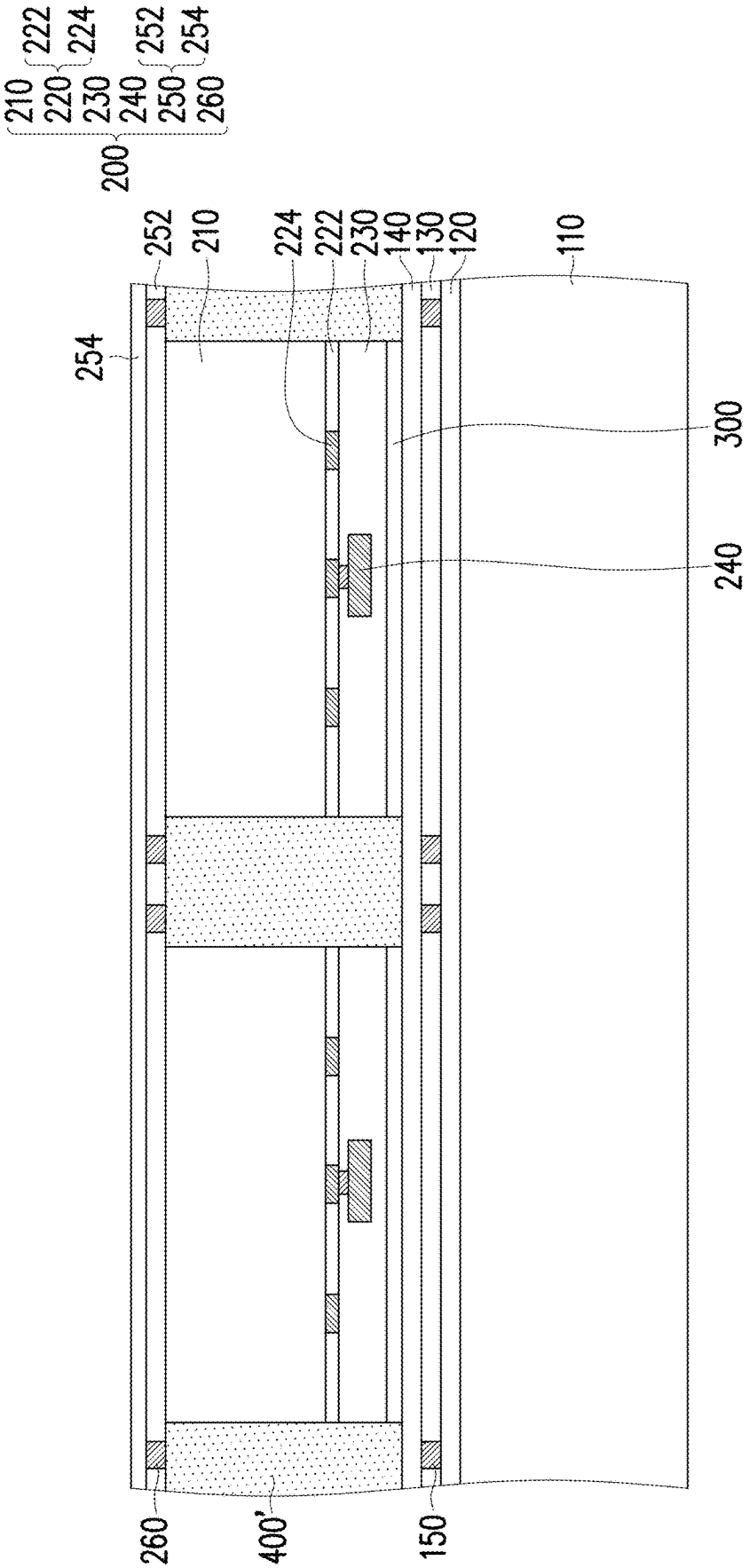
230
240
400'
224
222
230
140
130
120
110
300
210
240
150
$H_{200}$

FIG. 2B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210
220
230
240
250
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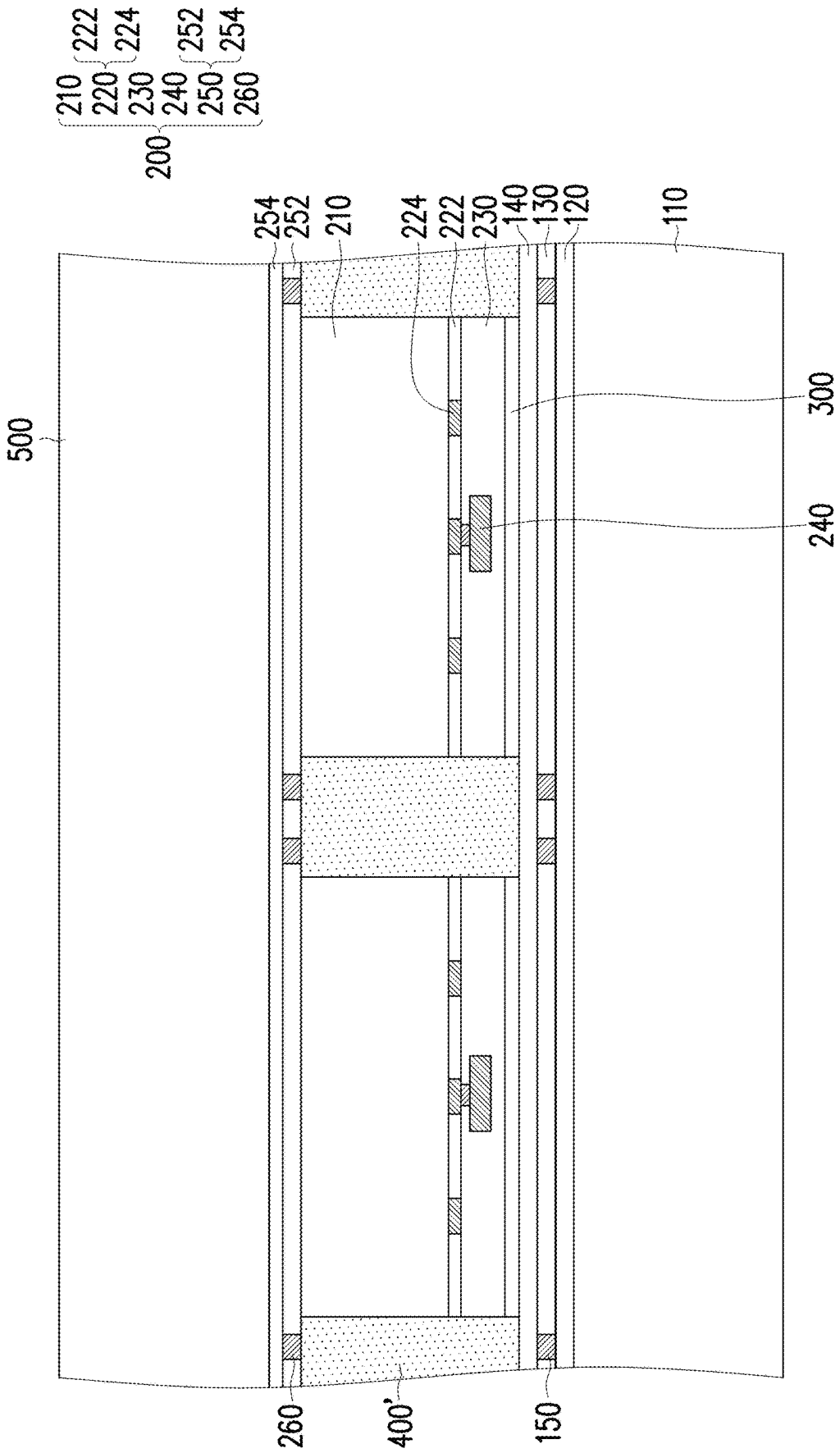
200
252
210
224
222
230
140
130
120
110
254
300
240
260
400'
150

FIG. 2C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222
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140
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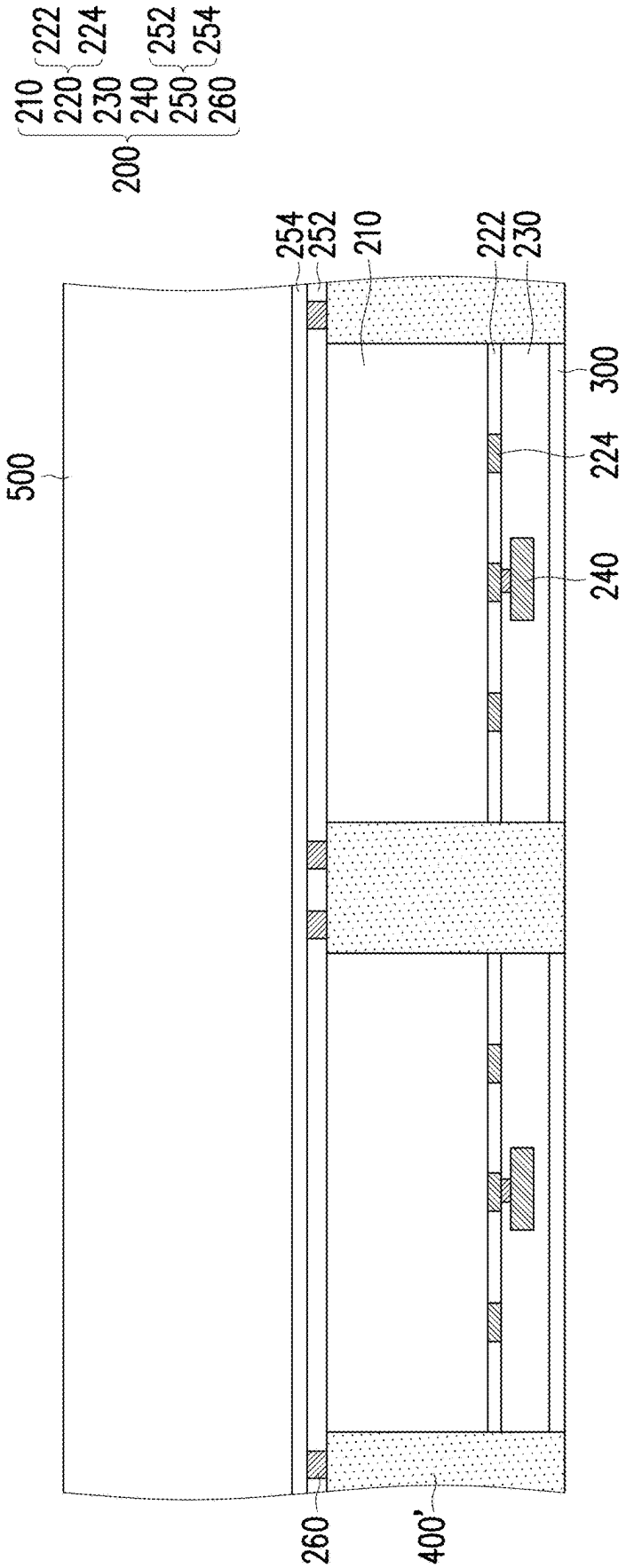
500
300
240
260
400'
150

FIG. 2D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252
254
210
220
230
240
250
260
200
254
252
210
222
230
300
500
224
240
260
400'

FIG. 2E

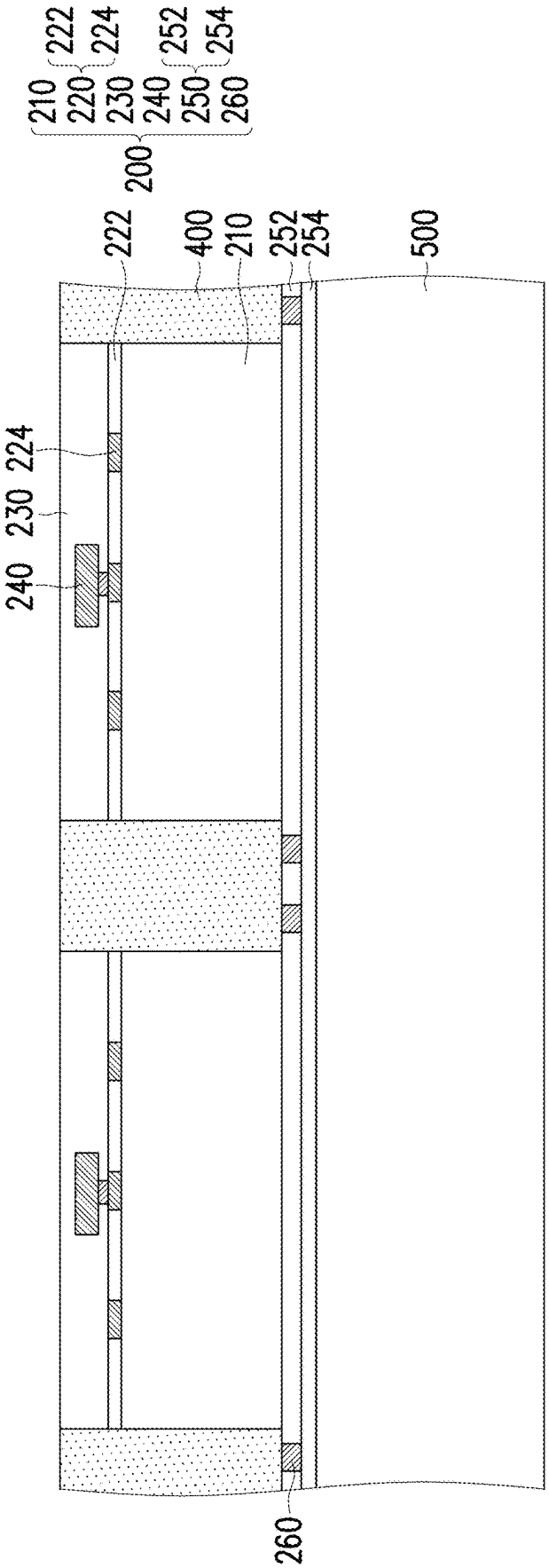
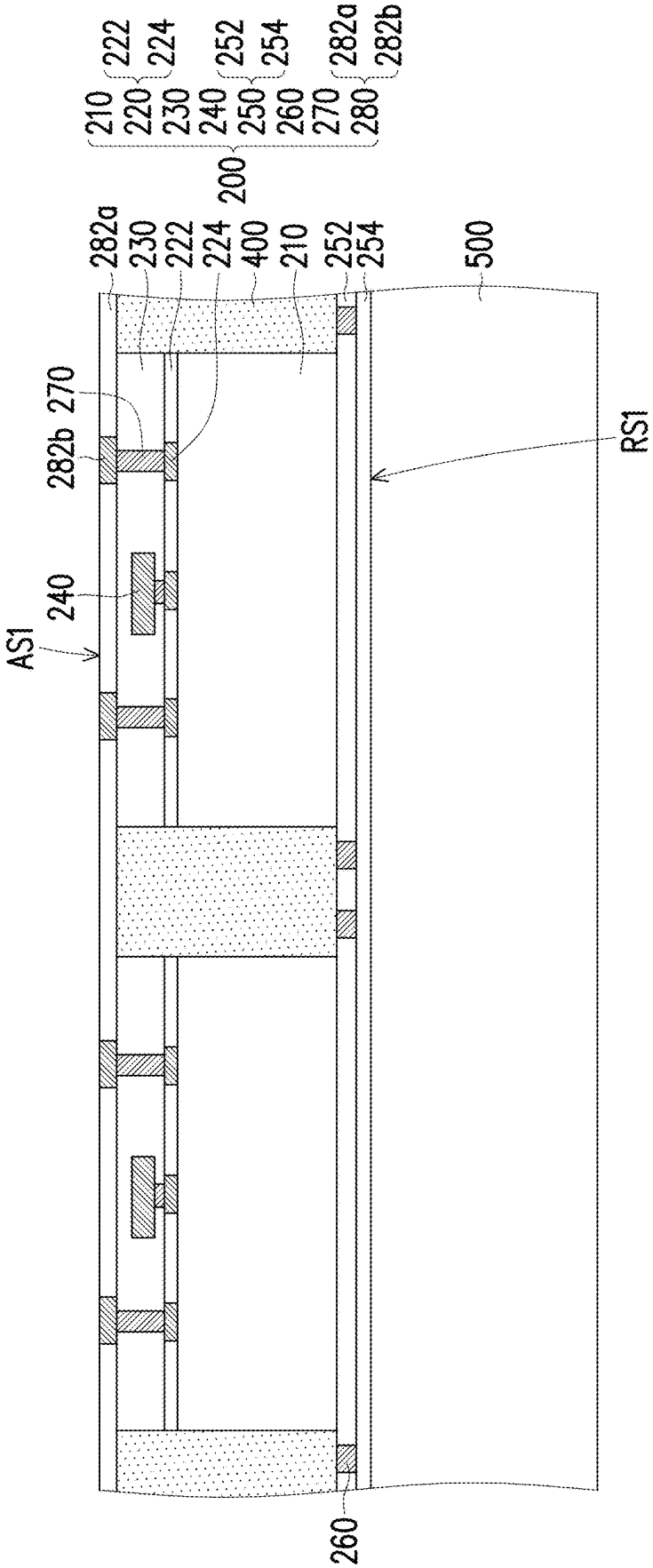

222
224
252
254
210
220
230
240
250
260
200
222
400
210
252
254
500
224
230
240
260

FIG. 2F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224
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254
282a
282b
210
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230
240
250
260
270
280
200
282a
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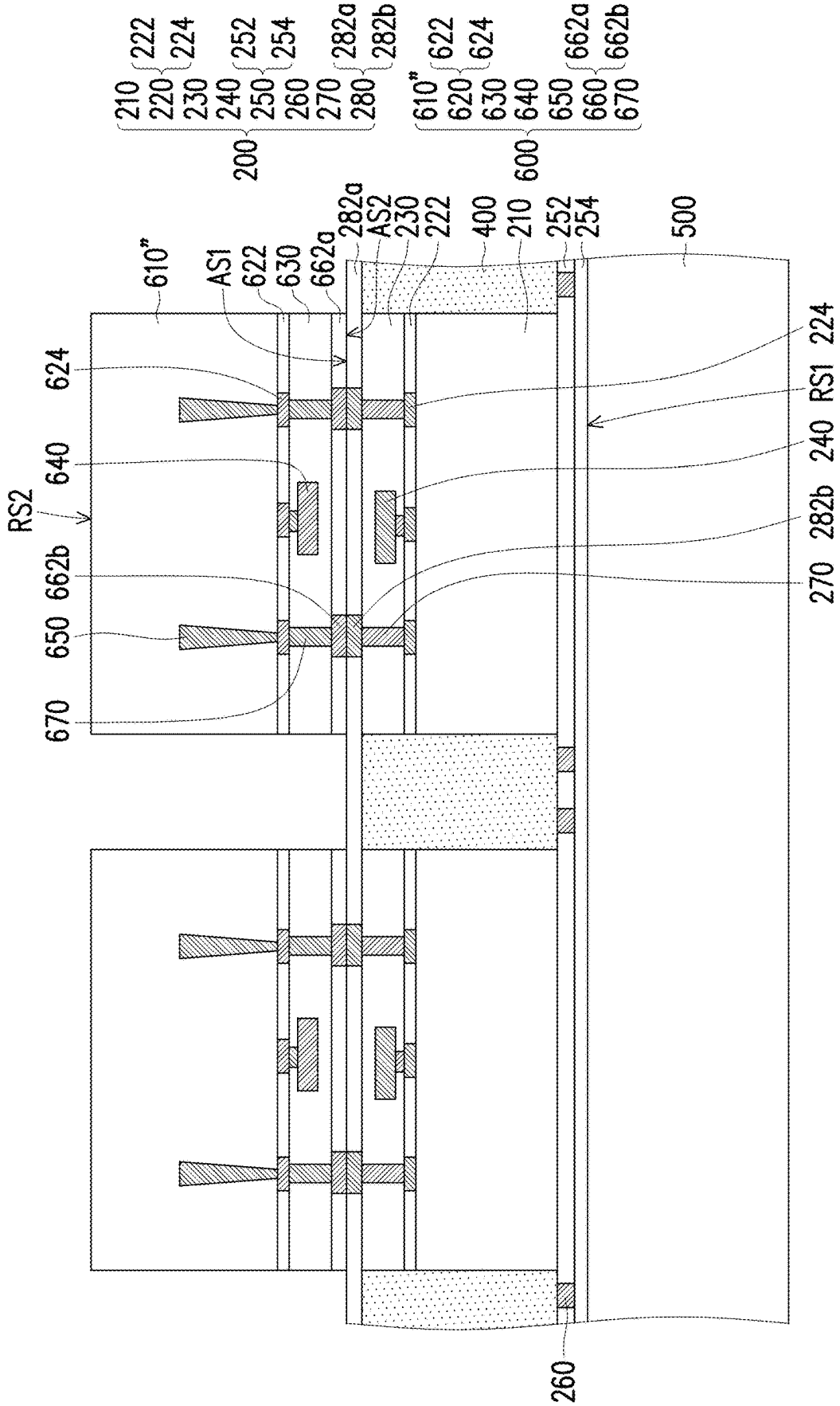
400
210
252
254
500
270
282b
RS1
240
AS1
260

FIG. 2G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270
280
282a
282b
600
610"
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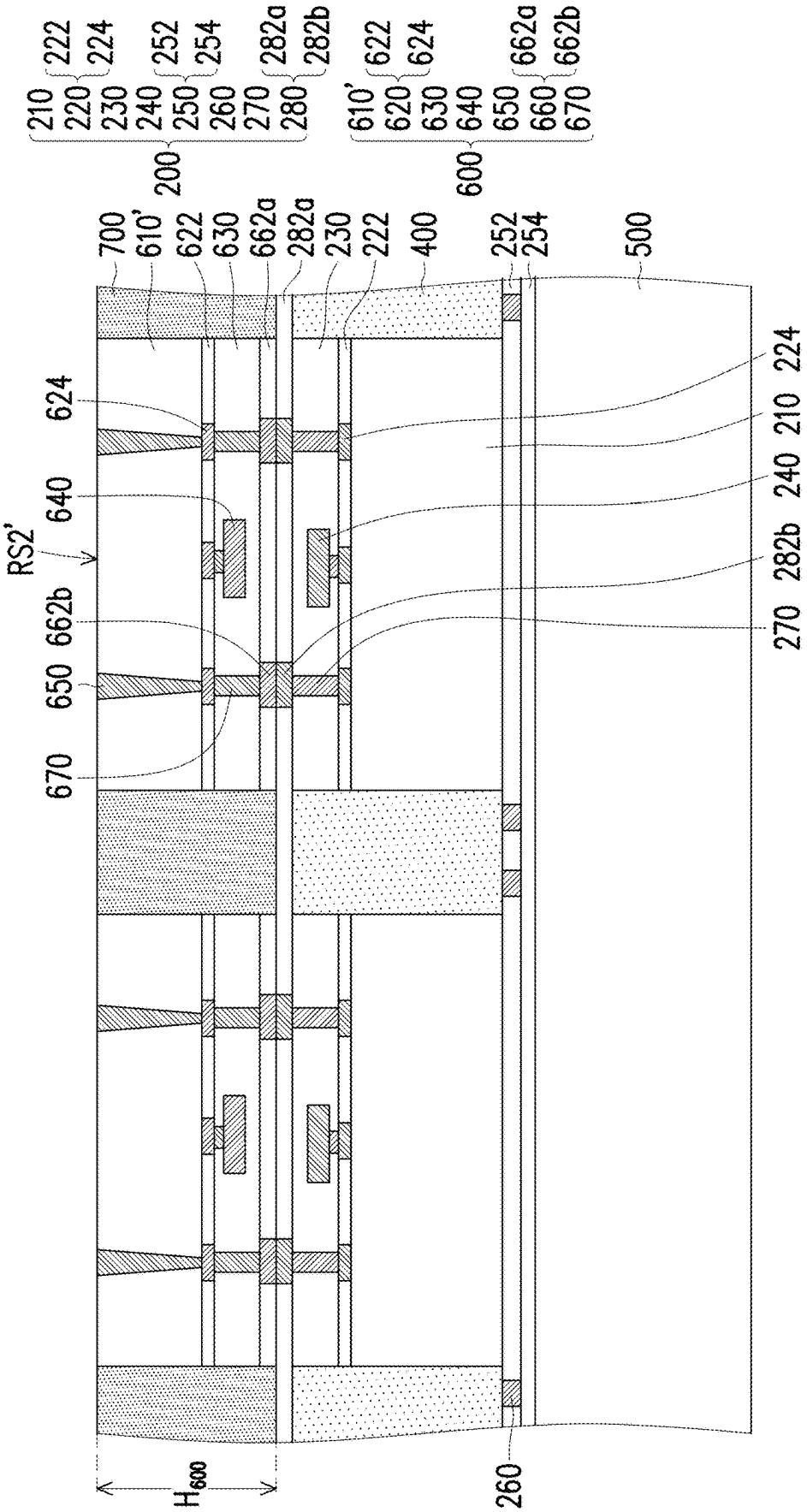
622
624
630
640
650
660
662a
662b
670
AS1
AS2
400
500
RS1
RS2

FIG. 2H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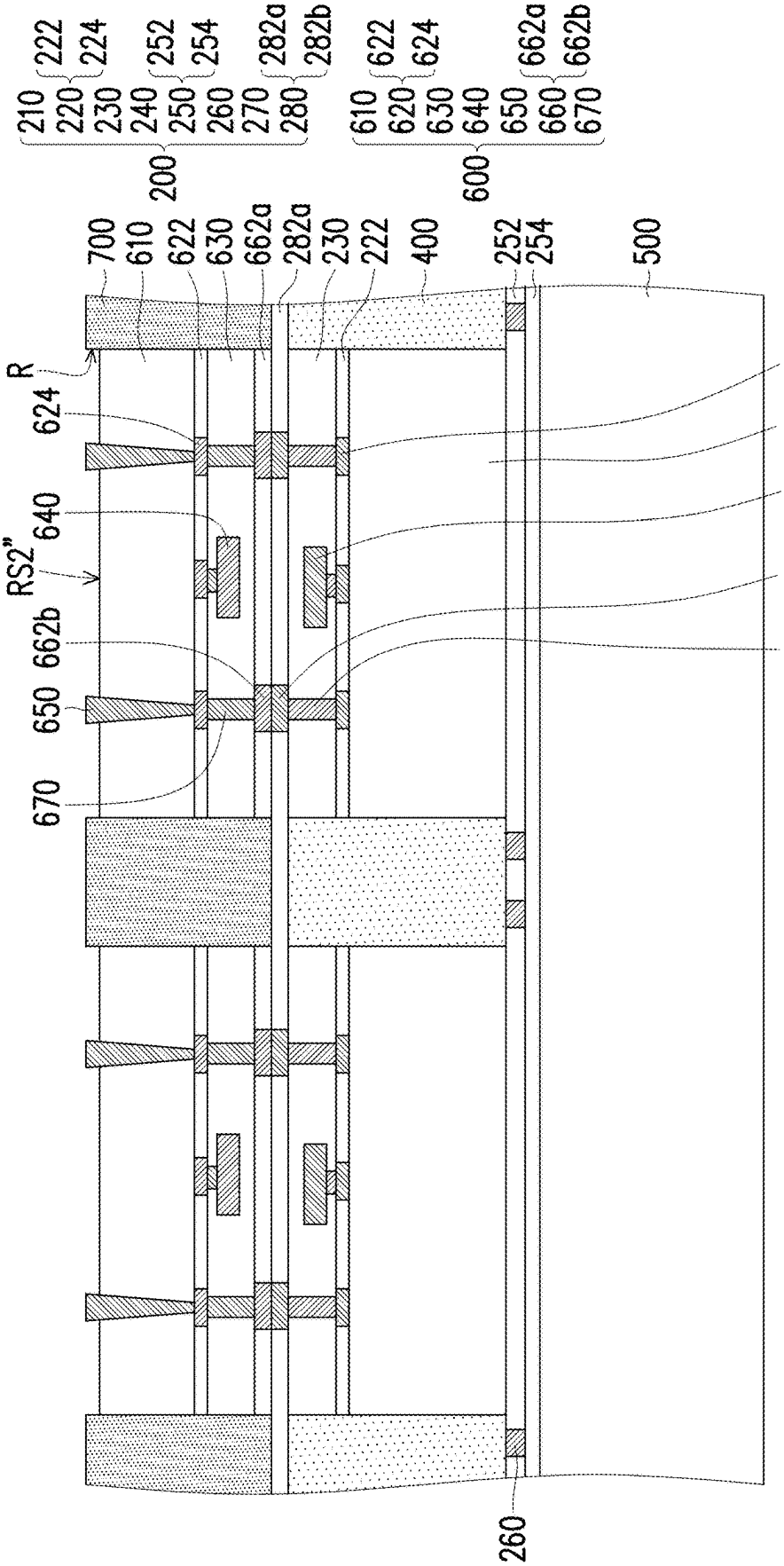
224
230
240
252
250
254
260
270
282a
280
282b
200
610'
620
622
624
630
640
650
660
662a
662b
670
600
700
610'
622
630
662a
282a
230
222
400
252
254
500
624
640
RS2'
662b
650
670
224
210
240
282b
270
$H_{600}$
260

FIG. 2I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250
254
260
270
282a
280
282b
200
610
620
622
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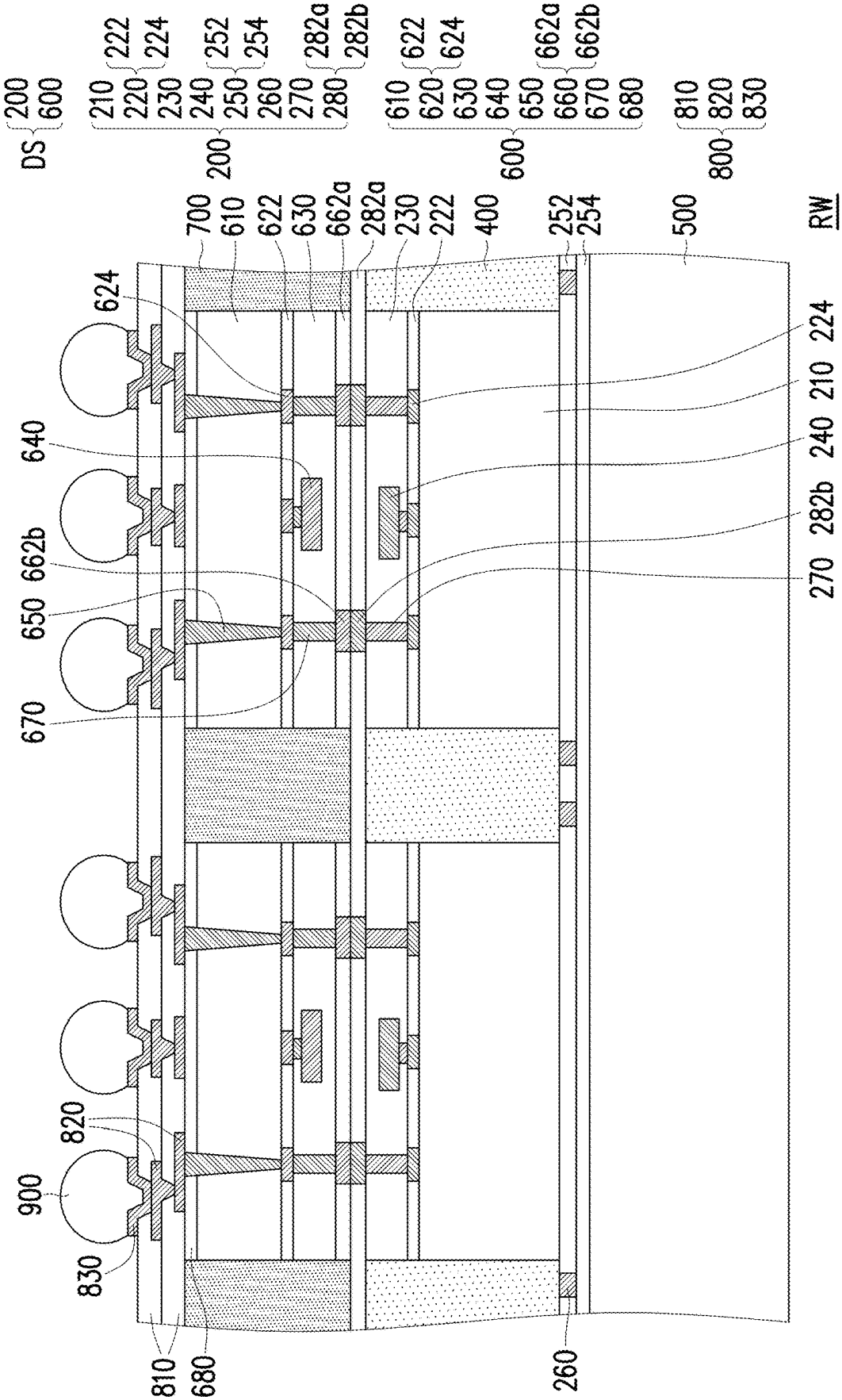
640
650
660
662a
662b
670
600
700
610
622
630
662a
282a
230
222
400
252
254
500
R
624
640
RS2"
662b
650
670
224
210
240
282b
270
260

FIG. 2J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240
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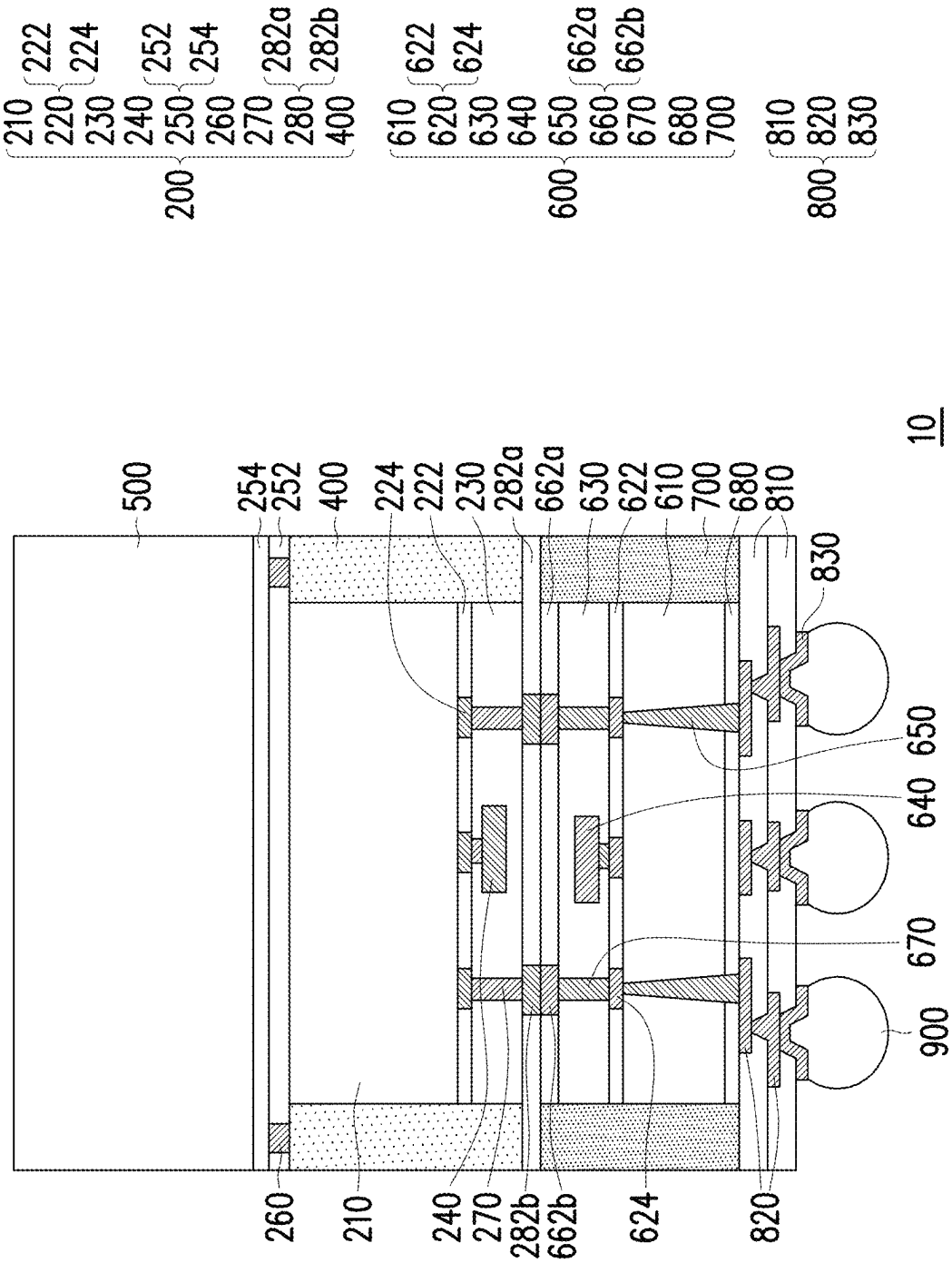
260
270
280
222
224
252
254
282a
282b
610
620
630
640
650
660
670
680
662a
662b
810
820
830
800
700
400
500
RW
624
640
662b
650
670
820
900
830
810
680
260
224
210
240
282b
270

FIG. 2K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210
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224
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240
250
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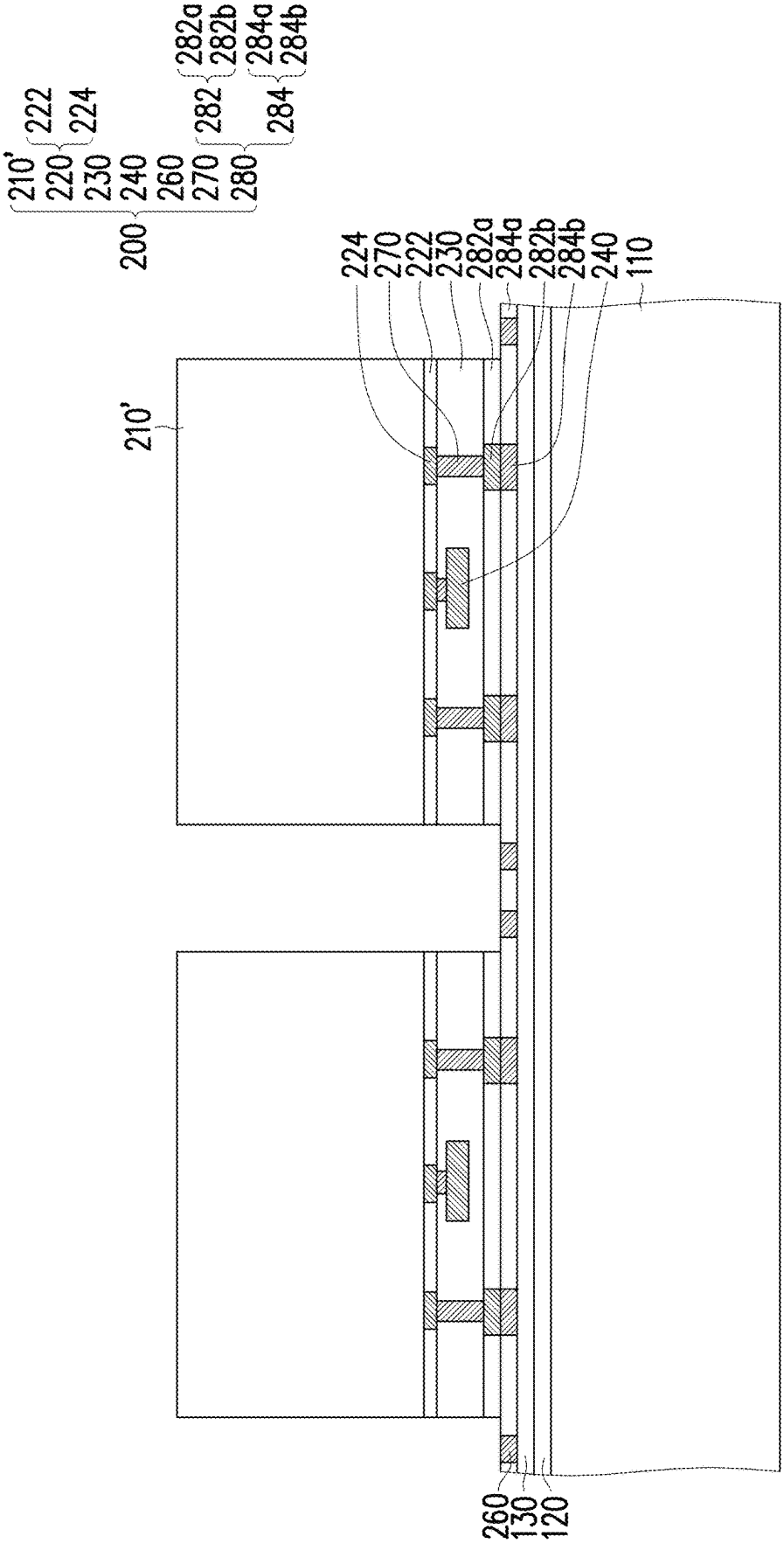
254
260
270
280
282a
282b
400
600
610
620
622
624
630
640
650
660
662a
662b
670
680
700
800
810
820
830
900
500
10

FIG. 3


210'
220
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284
282a
282b
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284b
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222
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282b
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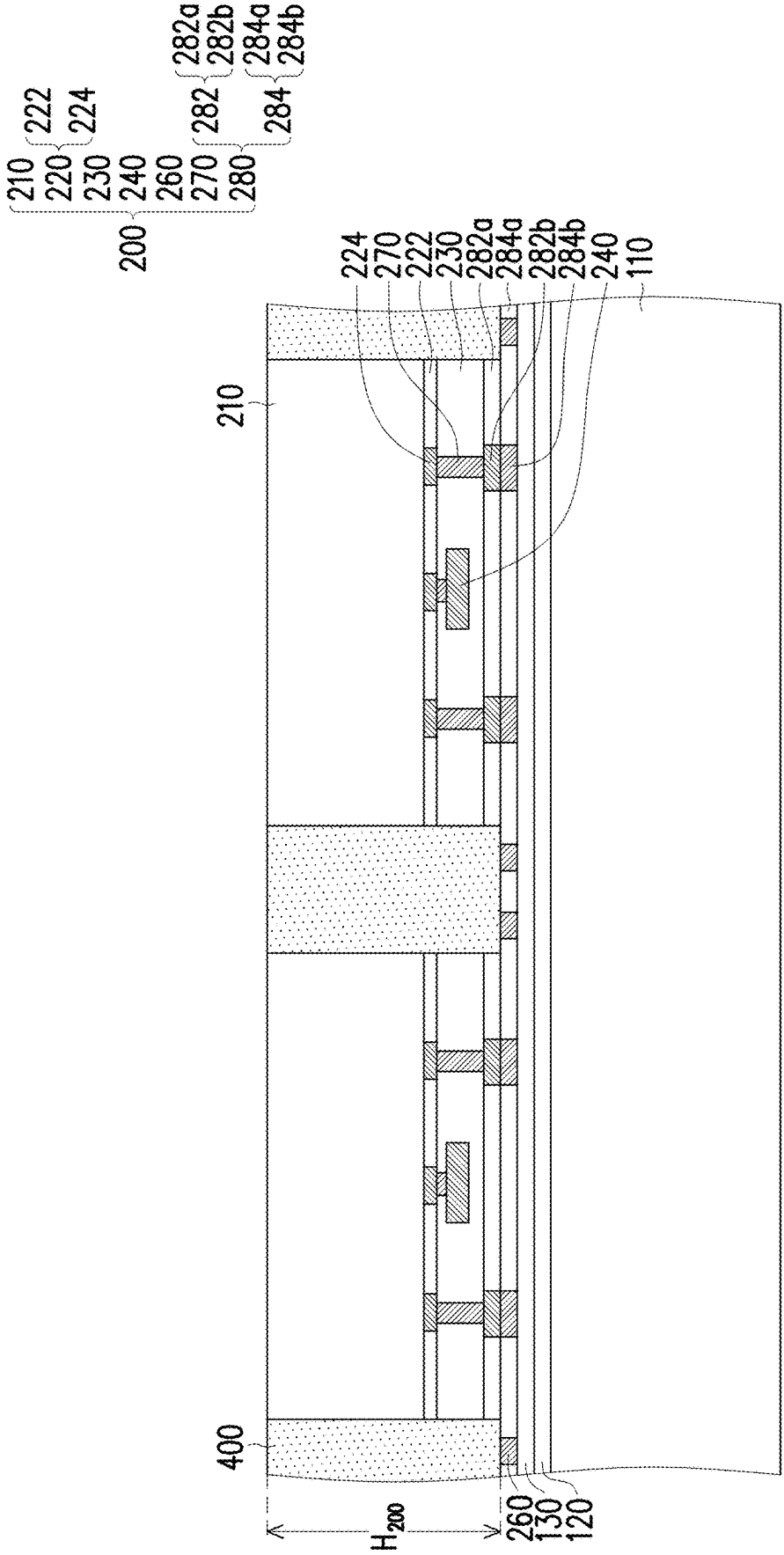
240
110
260
130
120

FIG. 4A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282a
282b
284
284a
284b
200
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230
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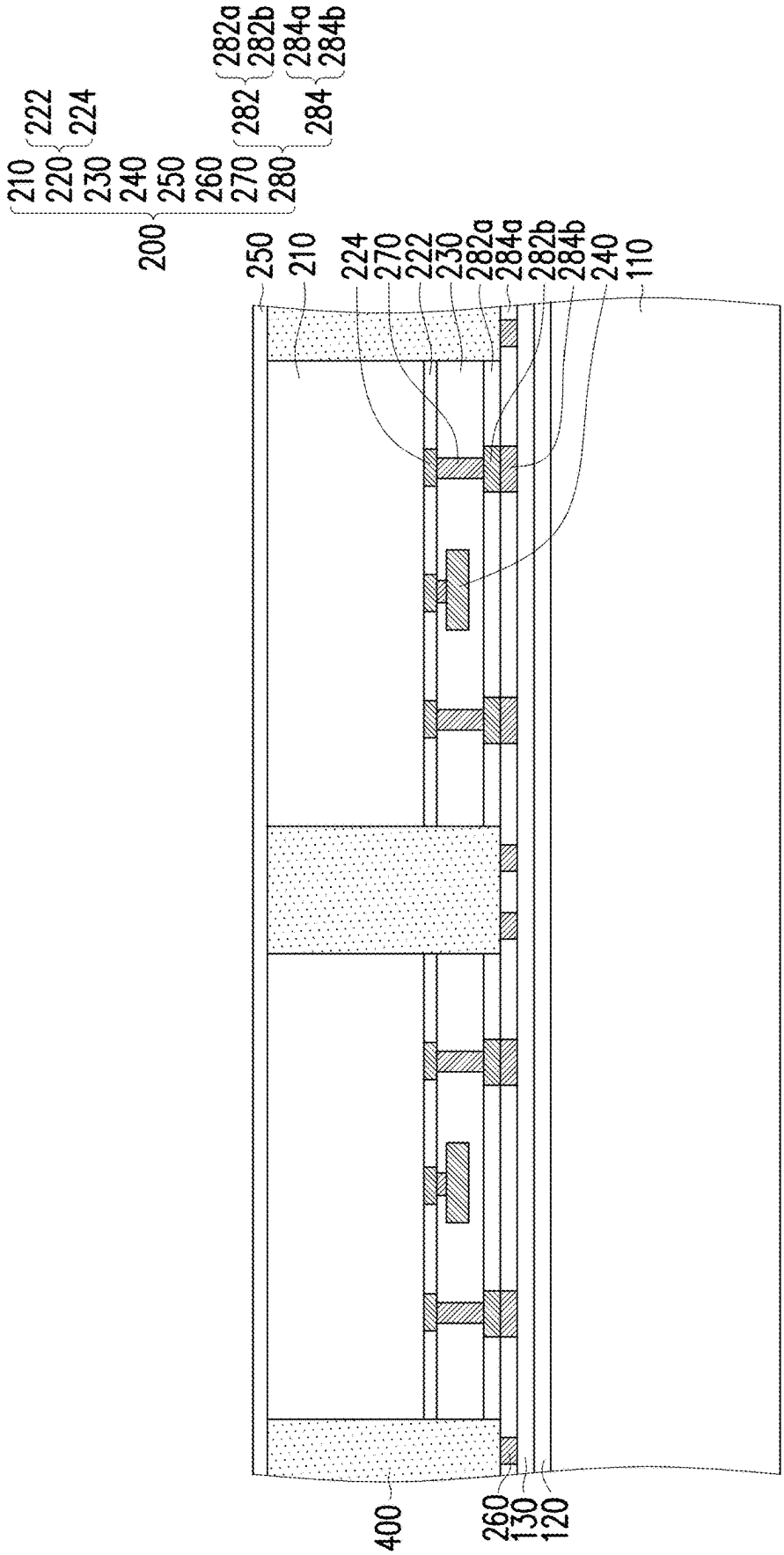
284a
282b
284b
240
110
210
400
$H_{200}$
260
130
120

FIG. 4B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282a
282b
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284a
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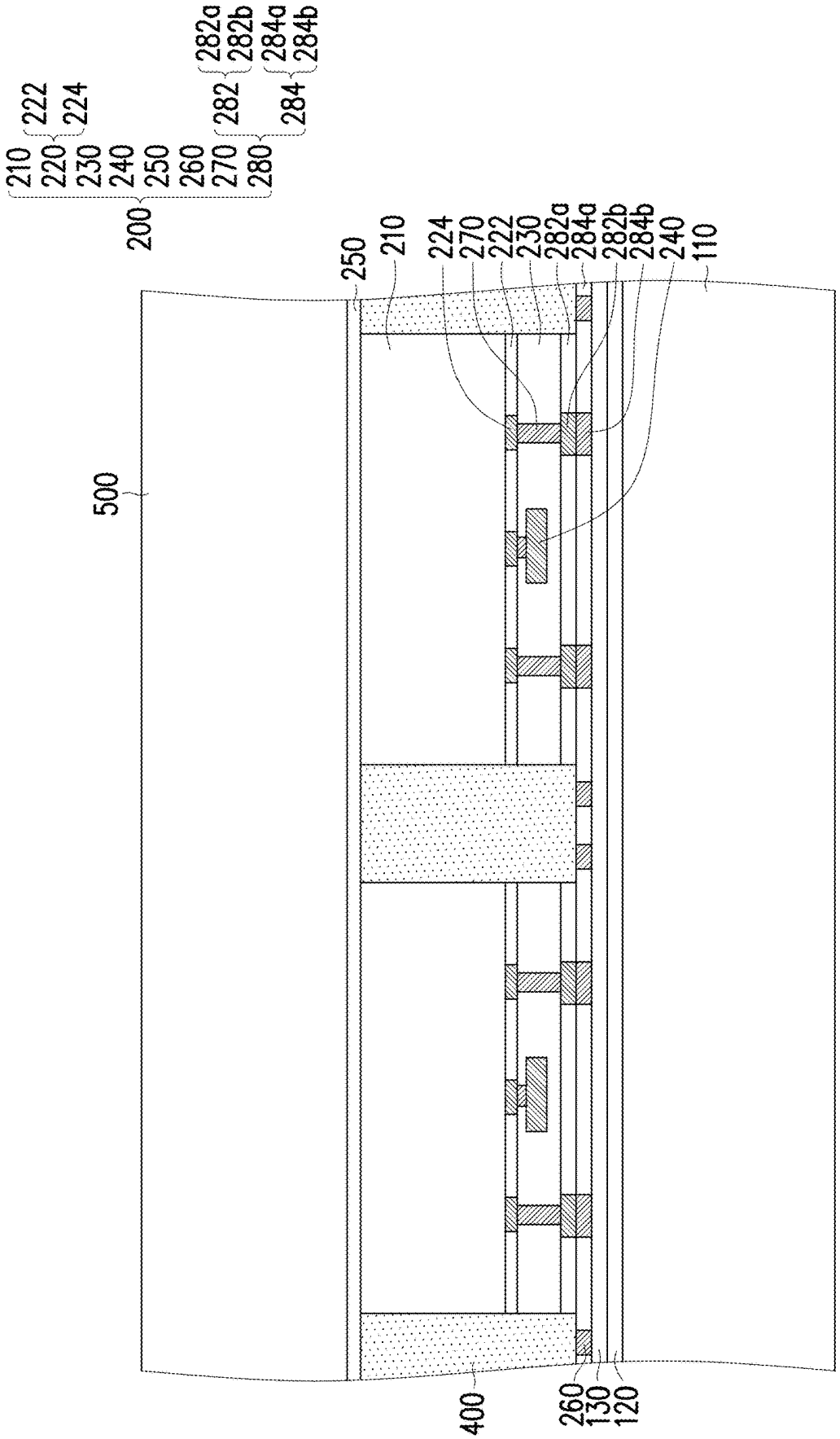
110
400
260
130
120

FIG. 4C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110
400
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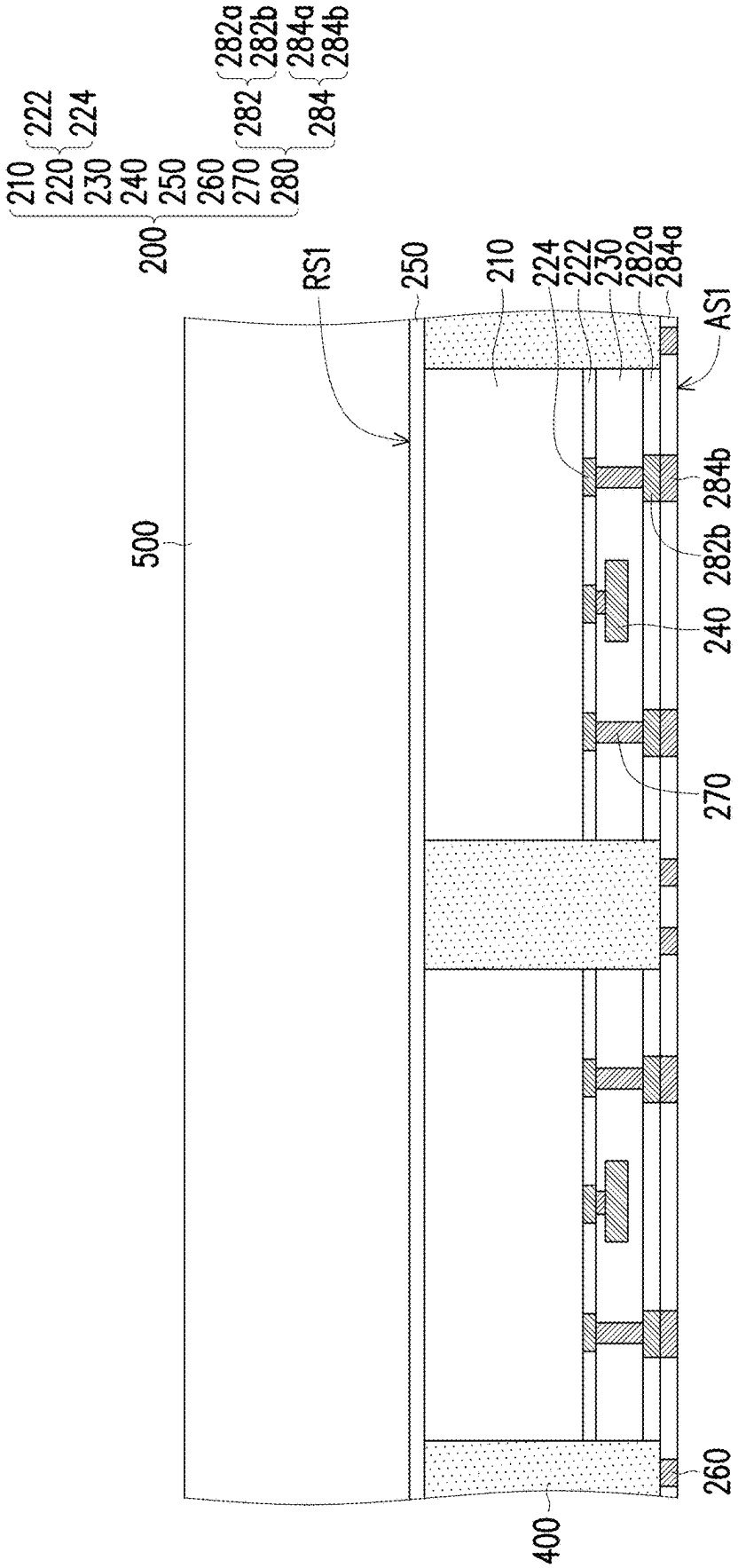
130
120

FIG. 4D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230
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AS1
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282b
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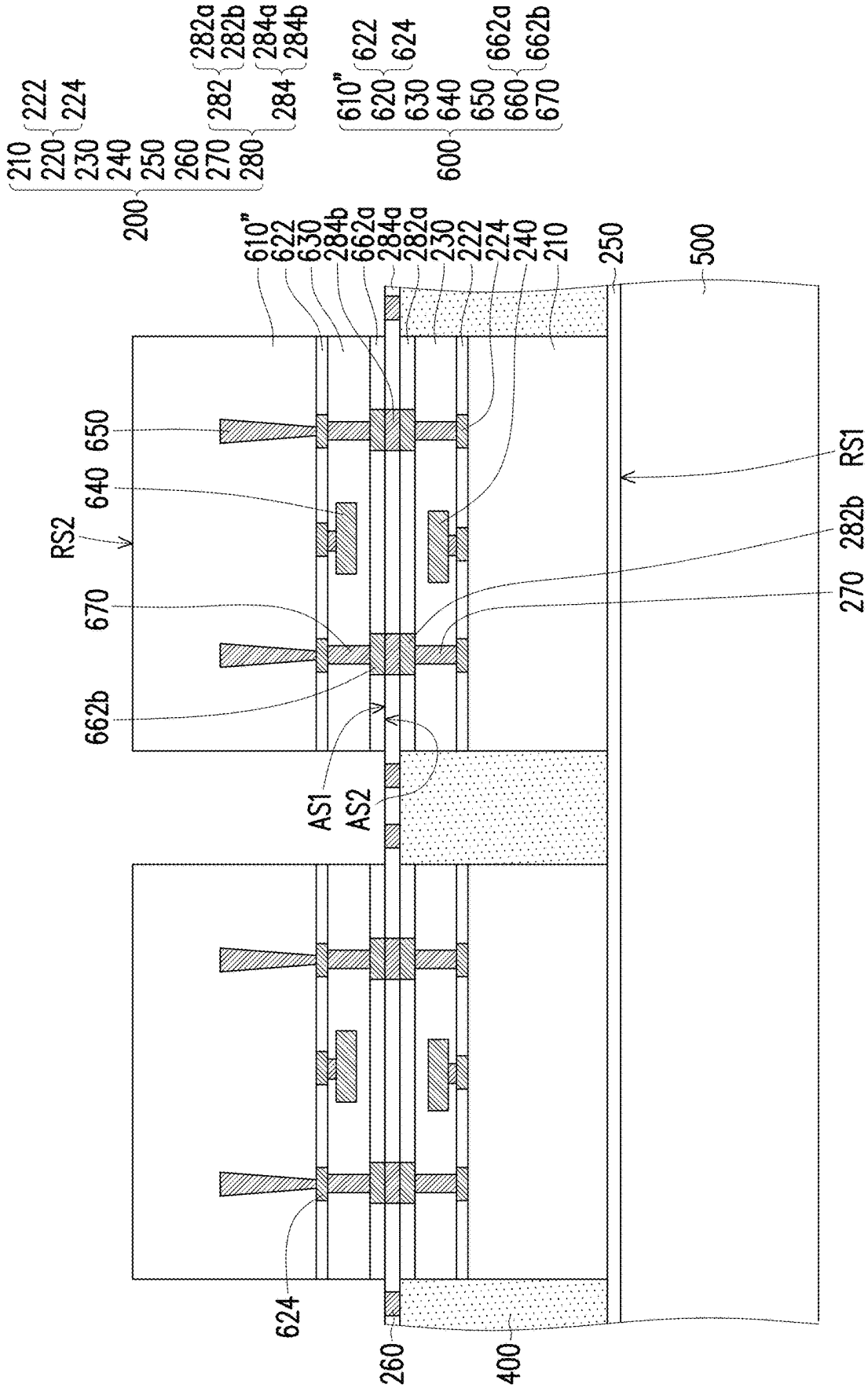
260
400

FIG. 4E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282b
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284a
284b
600
610"
620
622
624
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640
650
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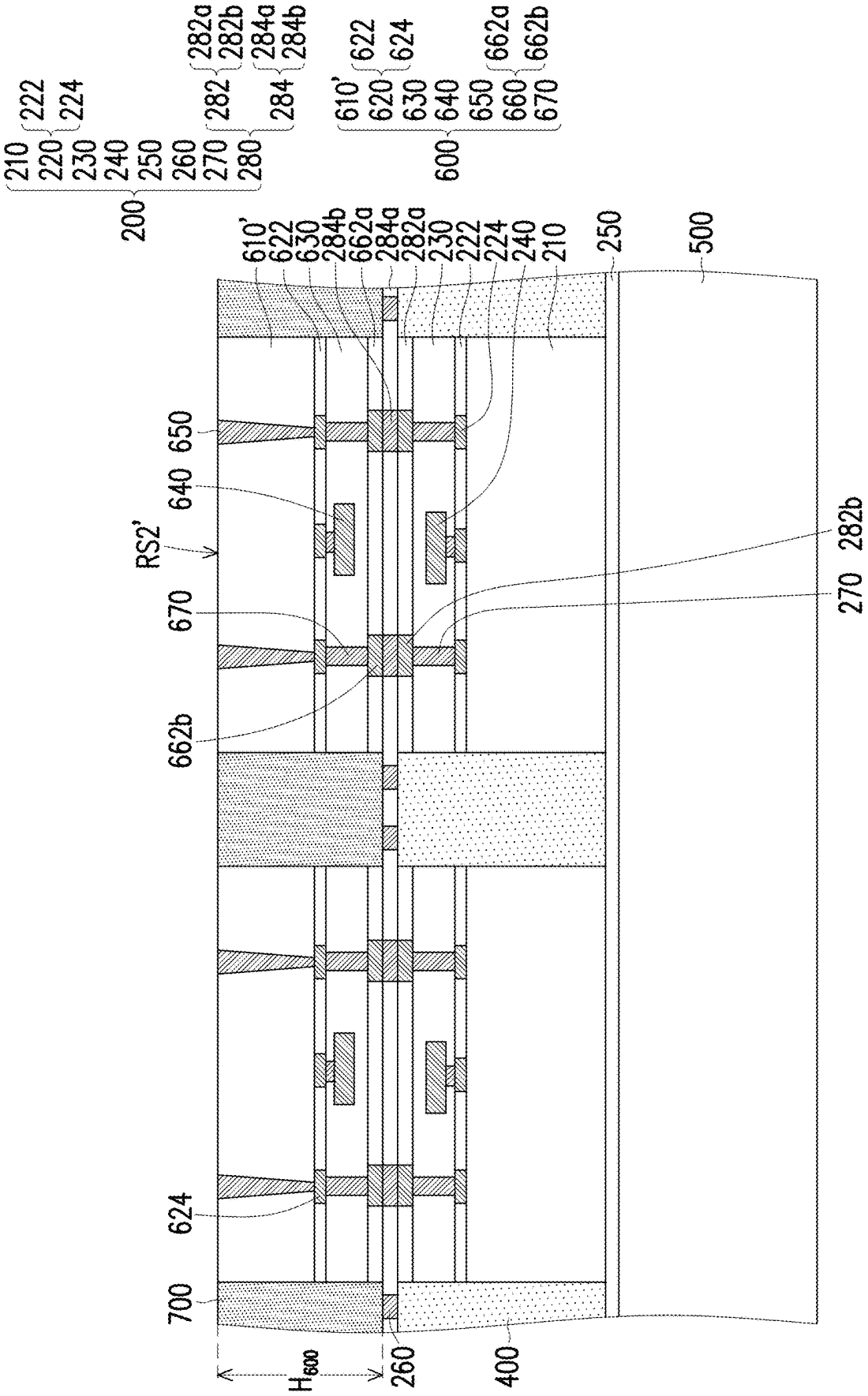
662a
662b
670
500
400
RS1
RS2
AS1
AS2

FIG. 4F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284a
284b
600
610'
620
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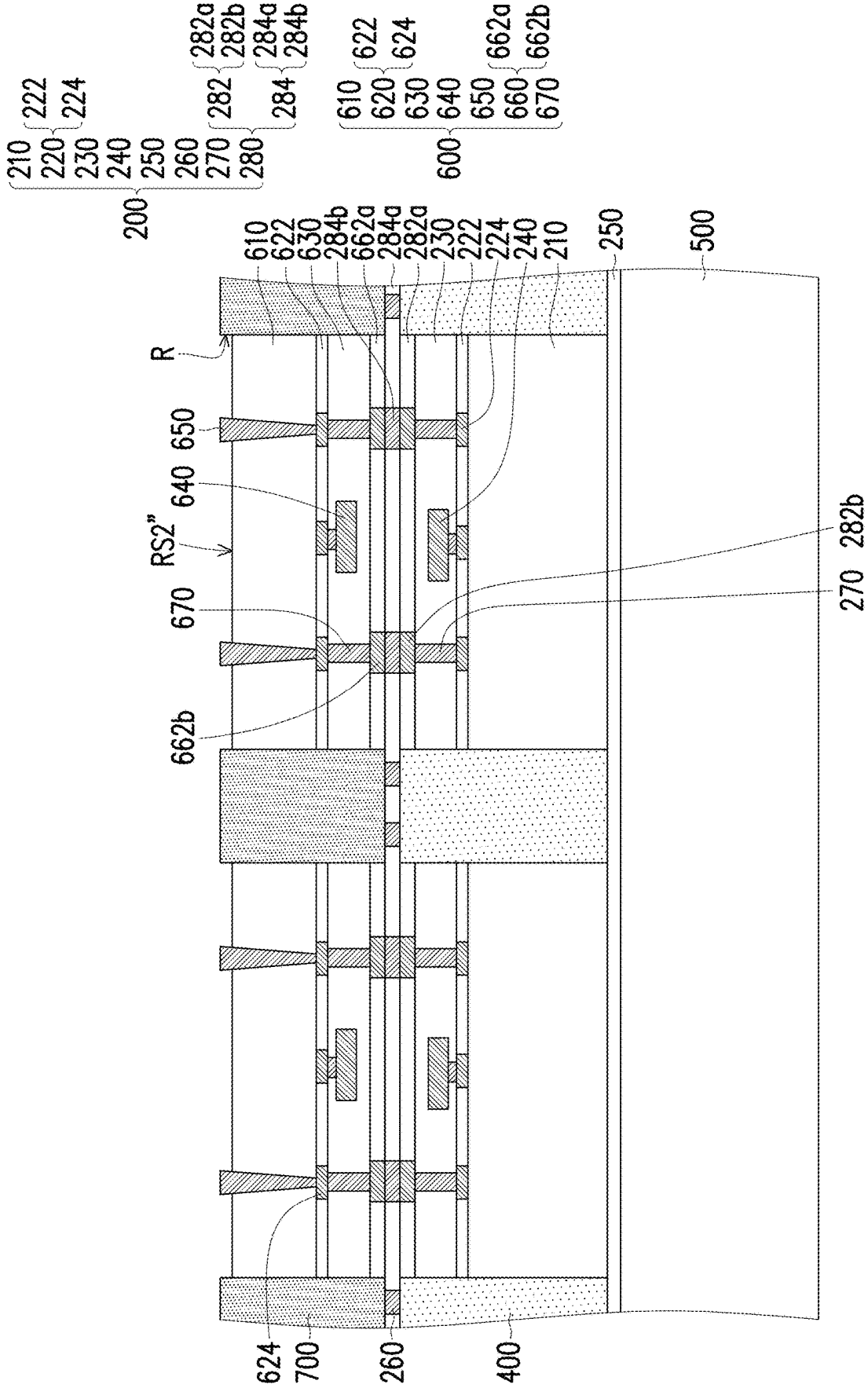
624
630
640
650
660
662a
662b
670
700
500
400
RS2'
$H_{600}$

FIG. 4G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284b
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284a
282a
230
222
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R
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RS2"
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282b
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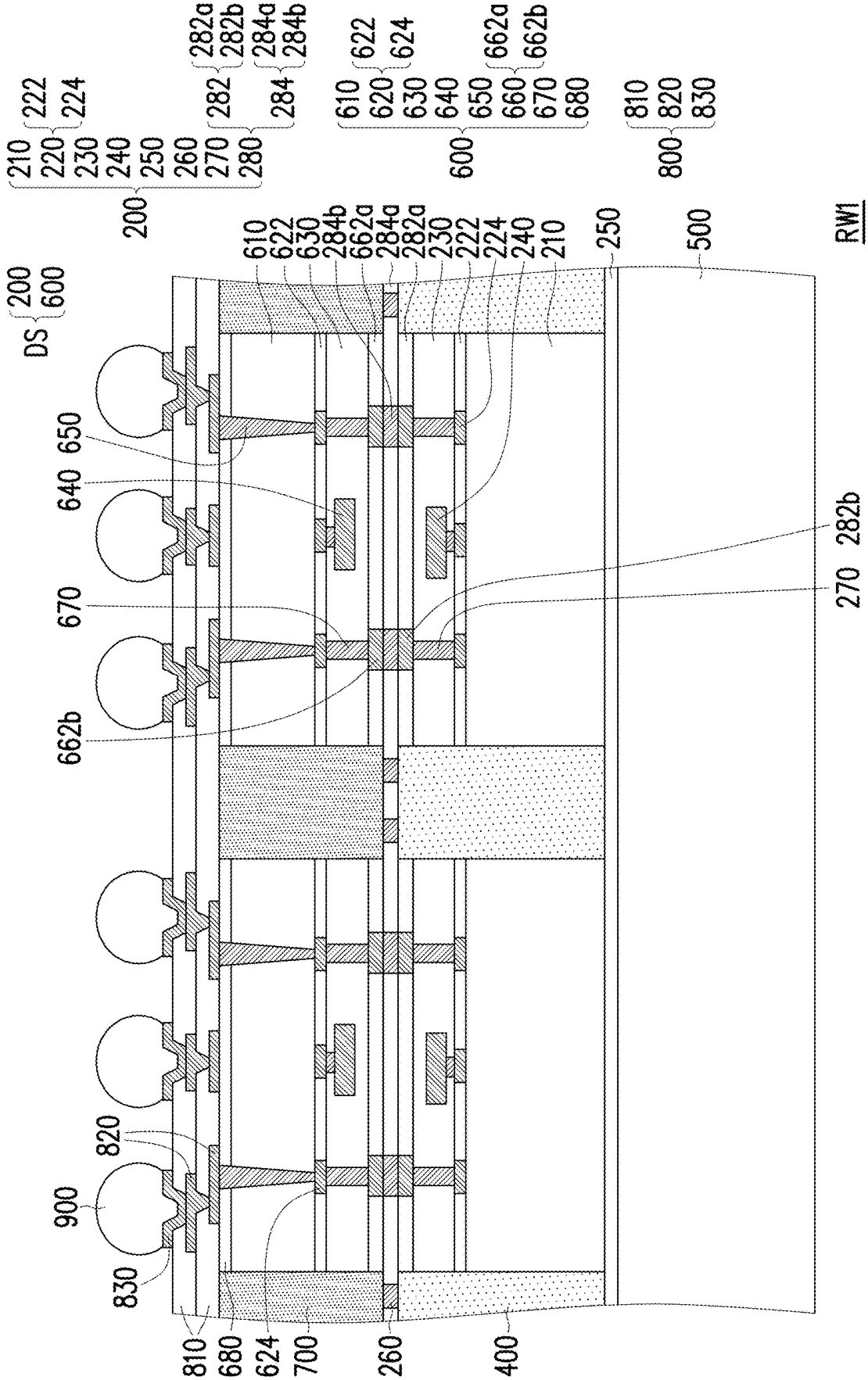
260
400

FIG. 4H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800
900
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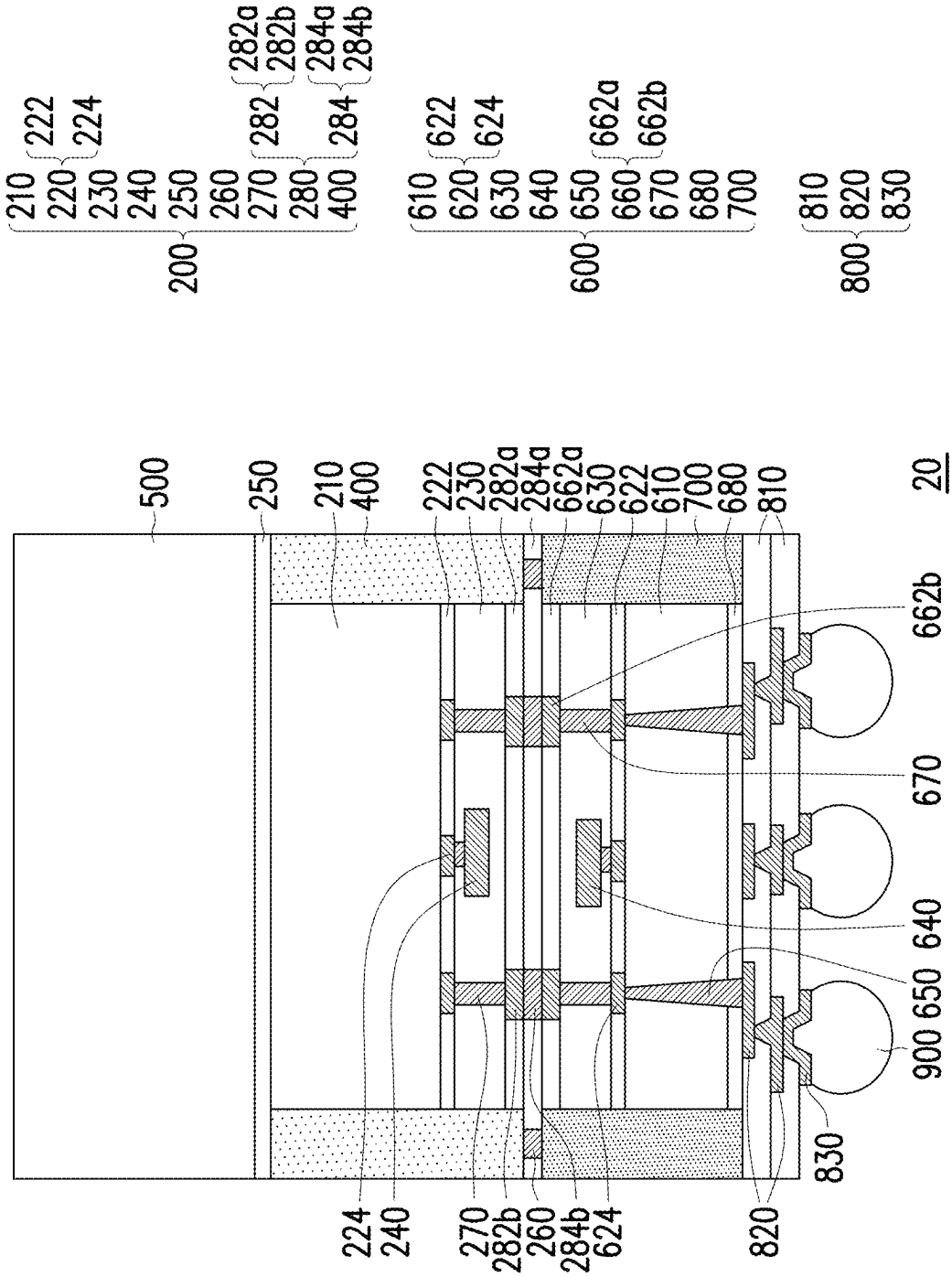
400
500
RW1

FIG. 4I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900
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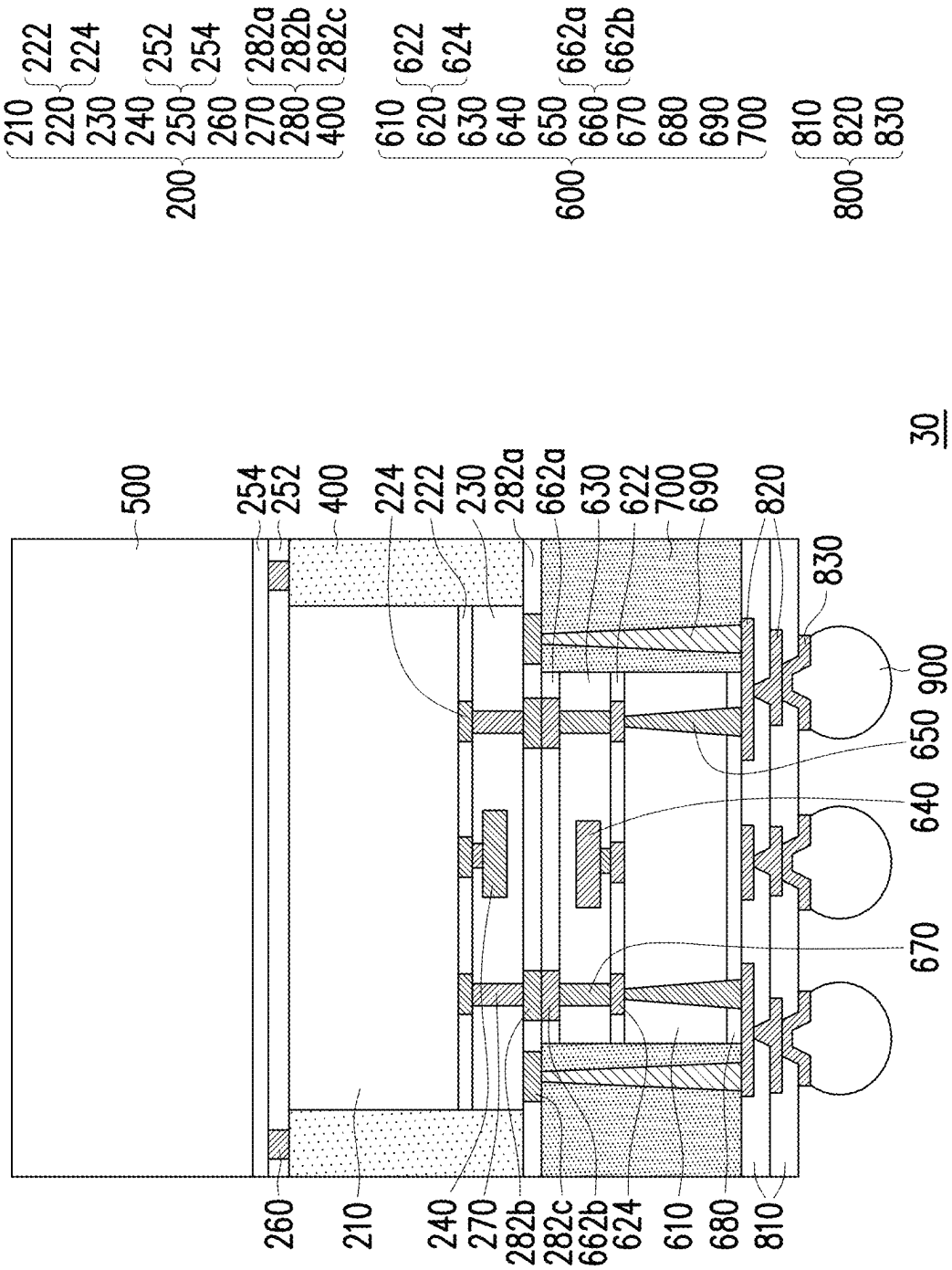
FIG. 5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282b
282c
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620
622
624
630
640
650
660
662a
662b
670
680
690
700
800
810
820
830
900
500
30

FIG. 6

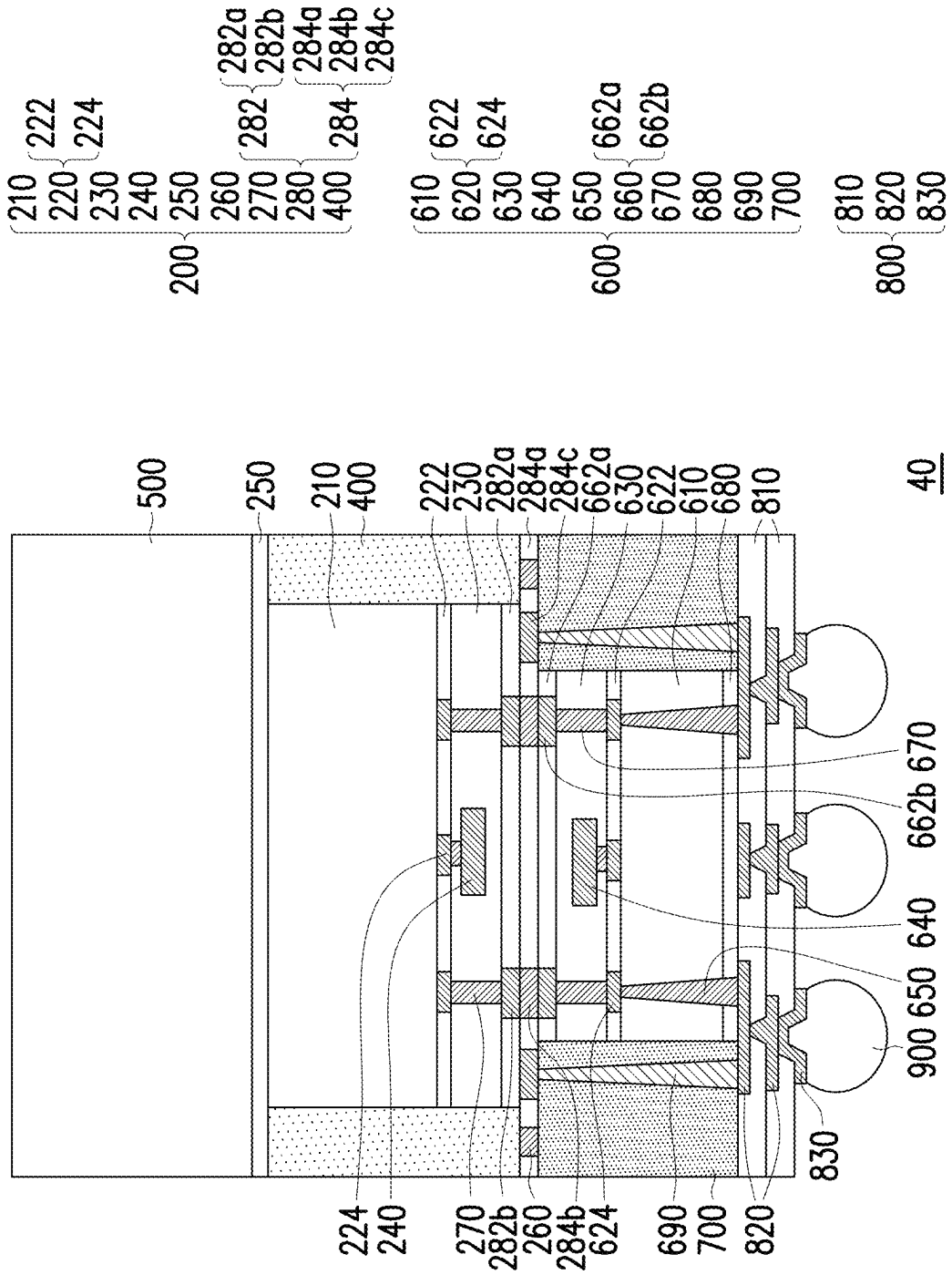
200
210
220
222
224
230
240
250
260
270
280
282
282a
282b
284
284a
284b
284c
400
600
610
620
622
624
630
640
650
660
662a
662b
670
680
690
700
800
810
820
830
900
500
40

FIG. 7

# PACKAGE

## CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation application of and claims the priority benefit of a prior application Ser. No. 17/344,928, filed on Jun. 10, 2021, now allowed. The prior application Ser. No. 17/344,928 is a divisional application of and claims the priority benefit of a prior application Ser. No. 16/572,622, filed on Sep. 17, 2019. The entirety of each of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

## BACKGROUND

Semiconductor devices and integrated circuits used in a variety of electronic apparatus, such as cell phones and other mobile electronic equipment, are typically manufactured on a single semiconductor wafer. The dies of the wafer may be processed and packaged with other semiconductor devices or dies at the wafer level, and various technologies and applications have been developed for the wafer level packaging. Integration of multiple semiconductor devices have become a challenge in the field.

## BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. **1** is a schematic view of a reconstructed wafer.

FIG. **2**A to FIG. **2**K are schematic cross-sectional views illustrating a manufacturing process of a reconstructed wafer in accordance with some embodiments of the disclosure.

FIG. **3** is a schematic cross-sectional view illustrating a package in accordance with some embodiments of the disclosure.

FIG. **4**A to FIG. **4**I are schematic cross-sectional view illustrating a manufacturing process of a reconstructed wafer in accordance with some alternative embodiments of the disclosure.

FIG. **5** is a schematic cross-sectional view illustrating a package in accordance with some alternative embodiments of the disclosure.

FIG. **6** is a schematic cross-sectional view illustrating a package in accordance with some alternative embodiments of the disclosure.

FIG. **7** is a schematic cross-sectional view illustrating a package in accordance with some alternative embodiments of the disclosure.

## DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in



which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as "beneath," "below," "lower," "above," "upper" and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Other features and processes may also be included. For example, testing structures may be included to aid in the verification testing of the 3D packaging or 3DIC devices. The testing structures may include, for example, test pads formed in a redistribution layer or on a substrate that allows the testing of the 3D packaging or 3DIC, the use of probes and/or probe cards, and the like. The verification testing may be performed on intermediate structures as well as the final structure. Additionally, the structures and methods disclosed herein may be used in conjunction with testing methodologies that incorporate intermediate verification of known good dies to increase the yield and decrease costs.

FIG. **1** is a schematic view of a reconstructed wafer RW. Referring to FIG. **1**, the reconstructed wafer RW includes a plurality of die stacks DS arranged in an array. In some embodiments, the die stacks DS are assemblies of known good dies (KND). For example, dies formed from a wafer are inspected and tested. Subsequently, the dies that are being determined as good dies are picked-and-placed onto another wafer to form the reconstructed wafer RW. As such, the reconstructed wafer RW has a high yield (i.e. 100% good die). In some embodiments, the reconstructed wafer RW may undergo further processing, such as a dicing step, to form a plurality of packages. The manufacturing process of the reconstructed wafer RW will be described in detail below in conjunction with FIG. **2**A to FIG. **2**K.

FIG. **2**A to FIG. **2**K are schematic cross-sectional views illustrating a manufacturing process of a reconstructed wafer RW in accordance with some embodiments of the disclosure. Referring to FIG. **2**A, a carrier substrate **110** is provided. In some embodiments, the carrier substrate **110** includes semiconductor materials. For example, the carrier substrate **110** may be made of a suitable elemental semiconductor, such as crystalline silicon, diamond, or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. In some embodiments, the carrier substrate **110** is free of active components and passive components. In some embodiments, the carrier substrate **110** is also free of wire routings. For example, the carrier substrate **110** may be a blank substrate which purely functions as a supporting element without serving any signal transmission function.

As illustrated in FIG. **2**A, a dielectric layer **120**, a dielectric layer **130**, and a bonding layer **140** are sequentially

disposed on the carrier substrate **110**. In other words, the dielectric layer **130** is sandwiched between the dielectric layer **120** and the bonding layer **140**. In some embodiments, the dielectric layer **120** and the dielectric layer **130** may be formed by suitable fabrication techniques, such as vapor deposition, spin coating, atomic layer deposition (ALD), thermal oxidation, some other suitable deposition or growth process, or a combination thereof. The vapor deposition may include, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), some other suitable vapor deposition process, or a combination thereof. In some embodiments, materials of the dielectric layer **120** and the dielectric layer **130** may be the same. For example, the dielectric layer **120** and the dielectric layer **130** may be polyimide, polybenzoxazole (PBO), benzocyclobutene (BCB), a nitride such as silicon nitride, an oxide such as silicon oxide, undoped silicate glass (USG), phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a combination thereof, or the like. However, the disclosure is not limited thereto. In some alternative embodiments, the material of the dielectric layer **120** may be different from the material of the dielectric layer **130**. For example, the material of the dielectric layer **120** may include undoped silicate glass (USG) while the material of the dielectric layer **130** may include a nitride such as silicon nitride. In some embodiments, the dielectric layer **120** and the dielectric layer **130** are able to control the warpage during subsequent processes. For example, the dielectric layer **120** and the dielectric layer **130** are able to reduce the overall warpage of the later-formed reconstructed wafer or package.

As illustrated in FIG. **2**A, a plurality of alignment marks **150** are embedded in the dielectric layer **130**. In other words, the alignment marks **150** are formed on the carrier substrate **110**. In some embodiments, the alignment marks **150** may be a patterned copper layer or other suitable patterned metal layer. In some embodiments, the alignment marks **150** may be formed by electroplating or deposition. It should be noted that the shapes and numbers of the alignment marks **150** are not limited in the disclosure, and may be designated based on the demand and/or design layout. In some embodiments, a top surface of the dielectric layer **130** is substantially levelled with top surfaces of the alignment marks **150**. In some embodiments, the alignment marks **150** are electrically isolated from other components. In other words, the alignment marks **150** are electrically floating.

In some embodiments, the bonding layer **140** is a smooth layer having a continuous even surface and overlaid on the dielectric layer **130** and the alignment marks **150**. In some embodiments, a material of the bonding layer **140** may include silicon oxynitride (SiON), silicon oxide, silicon nitride or the like, and the bonding layer **140** may be formed by deposition or the like. In some embodiments, the bonding layer **140** has a substantially uniform and even thickness.

As illustrated in FIG. **2**A, a plurality of dies **200** is attached to the carrier substrate **110**. In some embodiments, each die **200** includes a semiconductor substrate **210'**, an interconnection structure **220**, a passivation layer **230**, and a conductive pad **240**. In some embodiments, the interconnection structure **220** is disposed on the semiconductor substrate **210'**. The semiconductor substrate **210'** may be made of a suitable elemental semiconductor, such as crystalline silicon, diamond, or germanium; a suitable compound semiconductor, such as gallium arsenide, silicon carbide, indium arsenide, or indium phosphide; or a suitable alloy semiconductor, such as silicon germanium carbide, gallium arsenic phosphide, or gallium indium phosphide. In some embodiments, the semiconductor substrate **210'** may include active components (e.g., transistors or the like) and/or passive components (e.g., resistors, capacitors, inductors, or the like) formed therein.

In some embodiments, the interconnection structure **220** includes an inter-dielectric layer **222** and a plurality of conductive patterns **224** embedded in the inter-dielectric layer **222**. In some embodiments, the conductive patterns **224** of the interconnection structure **220** are electrically connected to the active components and/or the passive components embedded in the semiconductor substrate **210'**. In some embodiments, a material of the inter-dielectric layers **222** includes polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzoxazole (PBO), combinations thereof, or other suitable dielectric materials. The inter-dielectric layers **222** may be formed by suitable fabrication techniques, such as spin-on coating, lamination, chemical vapor deposition (CVD), or the like. In some embodiments, a material of the conductive patterns **224** includes aluminum, titanium, copper, nickel, tungsten, and/or alloys thereof. The conductive patterns **224** may be formed by, for example, electroplating, deposition, and/or photolithography and etching. For simplicity, the interconnection structure **220** is illustrated as having one layer of inter-dielectric layer **222** and one layer of conductive patterns **224** in FIG. **2**A. However, the disclosure is not limited thereto. In some alternative embodiments, the number of the layer of the inter-dielectric layer **222** and the number of the layer of the conductive patterns **224** may be adjusted depending on the routing requirements. For example, multiple layers of the inter-dielectric layer **122** and multiple layers of the conductive patterns **224** may be presented in the interconnection structure **220**, and the conductive patterns **224** and the inter-dielectric layers **222** may be stacked alternately.

In some embodiments, the conductive pad **240** is disposed over the interconnection structure **220**. In some embodiments, the conductive pad **240** is electrically connected to the conductive patterns **224** of the interconnection structure **200**. In some embodiments, the conductive pad **240** is used to establish electrical connection with other components (not shown) or dies (not shown) subsequently formed or provided. In some alternative embodiments, the conductive pad **240** may be test pads used to probe the die **200** in which the conductive pad **240** is included. In some embodiments, the conductive pad **240** may be aluminum pads, copper pads, or other suitable metal pads. It should be noted that the number and shape of the conductive pad **240** may be selected based on demand.

In some embodiments, the passivation layer **230** is formed over the interconnection structure **220** to seal the conductive pad **240**. In some embodiments, a material of the passivation layer **230** includes oxides, such as silicon oxide or the like. Alternatively, the passivation layer **230** may include polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzoxazole (PBO), or any other suitable polymer-based dielectric material. The passivation layer **230**, for example, may be formed by suitable fabrication techniques, such as spin-on coating, CVD, PECVD, or the like.

In some embodiments, the dies **200** may be capable of performing logic functions. For example, the dies **200** may be Central Process Unit (CPU) dies, Graphic Process Unit (GPU) dies, Field-Programmable Gate Array (FPGA), or the like.

In some embodiments, prior to the placement of the dies **200**, a bonding layer **300** is formed on each of the dies **200**.

For example, the bonding layer **300** is formed on the passivation layer **230** of the die **200**. In some embodiments, the bonding layer **300** is a smooth layer having a continuous even surface. In some embodiments, a material of the bonding layer **300** may include silicon oxynitride (SiON), silicon oxide, silicon nitride or the like, and the bonding layer **300** may be formed by deposition or the like.

In some embodiments, the dies **200** having the bonding layer **300** formed thereon are picked-and-placed onto the bonding layer **140** such that the bonding layer **300** is adhered to the bonding layer **140** through fusion bonding. The fusion bonding process may include a hydrophilic fusion bonding process, where a workable temperature is approximately greater than or substantially equal to about 100° C. and a workable pressure is approximately greater than or substantially equal to about 1 $kg/cm^2$. In some embodiments, the fusion bonding process does not involve metal to metal bonding. In some embodiments, the fusion bonded interface between the bonding layer **300** and the bonding layer **140** is substantially flat. For example, the fusion bonded interface has a roughness of less than 50 Å. In some embodiments, since the bonding layer **300** is fusion bonded to the bonding layer **140**, these bonding layers may be rather thin as compared to the conventional adhesive layers. For example, a thickness of the bonding layer **300** may range between 100 Å and 1 μm. Similarly, a thickness of the bonding layer **140** may also range between 100 Å and 1 μm. As illustrated in FIG. **2**A, the dies **200** are bonded to the carrier substrate **110** in a face down manner. That is, the interconnection structures **200** and the contact pads **240** of the dies **200** face the carrier substrate **110**. In some embodiments, the dies **200** are arranged in an array.

As illustrated in FIG. **2**A, the alignment marks **150** are arranged on a periphery region which surrounds a positioning location of the dies **200**. In the other words, the alignment marks **150** are disposed within a region aside of a location where the dies **200** are disposed on. With the presence of the alignment marks **150**, the transfer precision of the dies **200** onto the carrier substrate **110** may be effectively improved.

Referring to FIG. **2**B, an insulating layer **400'** is formed over the bonding layer **140** to laterally encapsulate the dies **200**. In some embodiments, a material of the insulating layer **400'** includes a molding compound, a polymeric material, such as polyimide, epoxy resin, acrylic resin, phenol resin, BCB, PBO, a combination thereof, or other suitable polymer-based dielectric materials. In some alternative embodiments, the insulating layer **400'** may include silicon oxide and/or silicon nitride. In some embodiments, the insulating layer **400'** further includes fillers. Alternatively, the insulating layer **400'** may be free of fillers.

In some embodiments, the insulating layer **400'** may be formed by the following steps. First, an insulating material (not shown) is formed over the bonding layer **140** to encapsulate the dies **200**. At this stage, the semiconductor substrates **210'** of the dies **200** are not revealed and are well protected by the insulating material. In some embodiments, the insulating material may be formed by a molding process (such as a compression molding process), a spin-coating process, a chemical vapor deposition (CVD) process, a plasma-enhanced chemical vapor deposition (PECVD) process, an atomic layer deposition (ALD) process, or the like. After the insulating material is formed, the insulating material is thinned until the semiconductor substrates **210'** of the dies **200** are exposed. In some embodiments, the semiconductor substrates **210'** and the insulating material are further thinned to reduce the overall thickness of the dies **200**. In some embodiments, the insulating material and the semiconductor substrates **210'** may be thinned or planarized through a grinding process, such as a mechanical grinding process, a CMP process, or the like. After the thinning process, each die **200** has a thinned semiconductor substrate **210** and the insulating layer **400'** is formed to expose the semiconductor substrate **210**. That is, top surfaces of the semiconductor substrates **210** are substantially coplanar with a top surface of the insulating layer **400'**. In some embodiments, the insulating layer **400'** may be referred to as "gap fill oxide." After the thinning process, each die **200** has a thickness $H_{200}$ of about 5 μm to about 100 μm. It should be noted that the foregoing process merely serves as an exemplary illustration, and the disclosure is not limited thereto. In some alternative embodiments, the insulating layer **400'** may be formed after the semiconductor substrates **210** are thinned.

Referring to FIG. **2**C, a bonding layer **250** is formed on the dies **200** and the insulating layer **400'** opposite to the carrier substrate **110**. In some embodiments, the bonding layer **250** includes a first sub-layer **252** and a second sub-layer **254** sequentially disposed on the semiconductor substrates **210** of the dies **200** and the insulating layer **400'**. For example, the first sub-layer **252** is attached to the insulating layer **400'** and the semiconductor substrate **210**. In some embodiments, the first sub-layer **252** may be formed by suitable fabrication techniques, such as vapor deposition, spin coating, atomic layer deposition (ALD), thermal oxidation, some other suitable deposition or growth process, or a combination thereof. The vapor deposition may include, for example, chemical vapor deposition (CVD), physical vapor deposition (PVD), some other suitable vapor deposition process, or a combination thereof. In some embodiments, a material of the first sub-layer **252** includes polyimide, polybenzoxazole (PBO), benzocyclobutene (BCB), a nitride such as silicon nitride, an oxide such as silicon oxide, undoped silicate glass (USG), phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), a combination thereof, or the like.

As illustrated in FIG. **2**C, a plurality of alignment marks **260** are embedded in the first sub-layer **252**. That is, the alignment marks **260** are embedded in the bonding layer **250**. The alignment marks **260** is similar to the alignment marks **150**, so the detailed description thereof is omitted herein. As illustrated in FIG. **2**C, the alignment marks **260** are arranged on a periphery region which surrounds the positioning location of the dies **200**. In the other words, the alignment marks **260** are disposed outside of the span of the dies **200**. In some embodiments, the alignment marks **260** are directly located on the insulating layer **400'**. In other words, the alignment marks **260** are directly in contact with the insulating layer **400'**.

In some embodiments, the second sub-layer **254** of the bonding layer **250** may be utilized for bonding purposes in the subsequent processes. In some embodiments, the second sub-layer **254** is a smooth layer having a continuous even surface and overlaid on the first sub-layer **252** and the alignment marks **260**. In some embodiments, a material of the second sub-layer **254** may include silicon oxynitride (SiON), silicon oxide, silicon nitride or the like, and the second sub-layer **254** may be formed by deposition or the like. In some embodiments, the second sub-layer **254** has a substantially uniform and even thickness. In some embodiments, a thickness of the second sub-layer **254** may range between 100 Å and 1 μm. In some embodiments, the bonding layer **250** and the alignment marks **260** may be considered as part of the dies **200**.

Referring to FIG. **2**D, the bonding layer **250** is attached to a carrier substrate **500** opposite to the carrier substrate **110**. That is, the dies **200** are bonded to the carrier substrate **500**, and the carrier substrates **110**, **500** are located on two opposite sides of the dies **200**. In some embodiments, the carrier substrate **500** is similar to the carrier substrate **110**, so the detailed description thereof is omitted herein. In some embodiments, the carrier substrate **500** is adhered to the bonding layer **250** through fusion bonding. The fusion bonding process may include a hydrophilic fusion bonding process, where a workable temperature is approximately greater than or substantially equal to about 100° C. and a workable pressure is approximately greater than or substantially equal to about 1 kg/cm$^2$. It should be noted that since the dies **200** are arranged on the carrier substrate **110** in a wafer form and the carrier substrate **500** is in wafer form as well, the bonding between the dies **200** and the carrier substrate **500** may be considered as a wafer-level process. That is, the bonding between the dies **200** and the carrier substrate **500** is a wafer-to-wafer bonding process. In some embodiments, with the presence of the alignment marks **260**, the wafer-to-wafer bonding precision may be effectively improved.

Referring to FIG. **2**D and FIG. **2**E, the carrier substrate **110**, the dielectric layer **120**, the dielectric layer **130**, the bonding layer **140**, and the alignment marks **150** are removed from the dies **200**, the bonding layer **300**, and the insulating layer **400'**. For example, as illustrated in FIG. **2**E, the insulating layer **400'** and the bonding layer **300** are exposed. In some embodiments, the carrier substrate **110**, the dielectric layer **120**, the dielectric layer **130**, the bonding layer **140**, and the alignment marks **150** are removed through a planarization process, an etching process, a stripping process, the like, or a combination thereof.

Referring to FIG. **2**E and FIG. **2**F, the bonding layer **300** and portions of the insulating layer **400'** are further removed to form an insulating layer **400** surrounding the semiconductor substrate **210**, the interconnection structure **220**, and the passivation layer **230**. For example, as illustrated in FIG. **2**F, the passivation layers **230** of the dies **200** are exposed. In some embodiments, the bonding layer **300** and portions of the insulating layer **400'** are removed through a planarization process, an etching process, the like, or a combination thereof. It should be noted that in some alternative embodiments, the step illustrated in FIG. **2**F is optional. That is, in some alternative embodiments, the subsequent processes may be performed without removing the bonding layer **300**.

Referring to FIG. **2**G, a plurality of bonding vias **270** and a bonding layer **280** are formed. In some embodiments, the bonding vias **270** are formed to penetrate through the passivation layer **230** to establish electrical connection with the conductive patterns **224** of the interconnection structure **220**. The bonding layer **280** is formed on the dies **200** and the insulating layer **400'**. For example, the bonding layer **280** is stacked on the passivation layer **230**, the bonding vias **270**, and the insulating layer **400**. In some embodiments, the bonding layer **280** is opposite to the bonding layer **250**. In some embodiments, the bonding layer **280** includes a dielectric layer **282***a* and a plurality of bonding pads **282***b* embedded in the dielectric layer **282***a*. In some embodiments, the bonding pads **282***b* of the bonding layer **280** are electrically connected to the bonding vias **270**. That is, the bonding vias **270** electrically connect the interconnection structure **220** and the bonding pads **282***b* of the bonding layer **280**.

In some embodiments, the bonding vias **270** and the bonding pads **282***b* may be formed via a dual damascene process. For example, the dielectric layer **282***a* is first formed on the passivation layer **230**. In some embodiments, a material of the dielectric layer **282***a* includes oxides, such as silicon oxide or the like. Alternatively, the dielectric layer **282***a* may include polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzoxazole (PBO), or any other suitable polymer-based dielectric material. The dielectric layer **282***a*, for example, may be formed by suitable fabrication techniques, such as spin-on coating, CVD, PECVD, or the like. Subsequently, trenches and via holes (not shown) are formed in the dielectric layer **282***a* and the passivation layer **230** by removing portions of theses layers. In some embodiments, a width of the trench is greater than a width of the via hole. Thereafter, a conductive material (not shown) is filled into the trenches and the via holes to form the bonding pads **282***b* and the bonding vias **270**, respectively. In some embodiments, the bonding vias **270** and the bonding pads **282***b* are formed by simultaneously filling via holes and overlying trenches (not shown). In some alternative embodiments, the bonding vias **270** may be formed before the dielectric layer **282***a* and the bonding pads **282***b*. In some embodiments, a width of each bonding pad **282***b* may be greater than a width of each underlying bonding via **270**. In some embodiments, the bonding vias **270** and the bonding pads **282***b* include the same material. Materials for the bonding vias **270** and the bonding pads **282***b* are, for example, aluminum, titanium, copper, nickel, tungsten, or alloys thereof. In some embodiments, the bonding vias **270** and the bonding layer **280** may be considered as part of the dies **200**.

In some embodiments, top surfaces of the bonding pads **282***b* and a top surface of the dielectric layer **282***a* may be collectively referred to as active surfaces AS**1** of the dies **200**. On the other hand, the surface of the bonding layer **250** facing the carrier substrate **500** may be referred to as rear surfaces RS**1** of the dies **200**. As shown in FIG. **2**G, the top surfaces of the bonding pads **282***b* and the top surface of the dielectric layer **282***a* are substantially located at the same level height to provide an appropriate active surfaces AS**1** for hybrid bonding. Although not illustrated, in some embodiments, some of the bonding vias **270** and some of the bonding pads **282***b* may be disposed directly above the conductive pads **240** to establish electrical connection between the conductive pads **240** and other elements. That is, in some embodiments, some of the conductive pads **240** are electrically floating while some of the conductive pads **240** are able to transmit signal.

Referring to FIG. **2**H, a plurality of dies **600** are provided. In some embodiments, each die **600** includes a semiconductor substrate **610"**, an interconnection structure **620**, a passivation layer **630**, a conductive pad **640**, a plurality of through semiconductor vias (TSV) **650**, a bonding layer **660**, and a plurality of bonding vias **670**. In some embodiments, the semiconductor substrates **610"** of the die **600** in FIG. **2**H is similar to the semiconductor substrate **210'** of the die **200** in FIG. **2**A, so the detailed descriptions thereof is omitted herein. As illustrated in FIG. **2**H, the interconnection structure **620** is disposed on the semiconductor substrate **610"**. In some embodiments, the interconnection structure **620** includes an inter-dielectric layer **622** and a plurality of conductive patterns **624**. The inter-dielectric layer **622** and the conductive patterns **624** of the interconnection structure **620** are respectively similar to the inter-dielectric layer **222** and the conductive patterns **224** of the interconnection structure **220**, so the detailed descriptions thereof are omitted herein.

In some embodiments, the conductive pad **640** is disposed over and electrically connected to the interconnection struc-

ture **620**. On the other hand, the passivation layer **630** is formed over the interconnection structure **620** to seal the conductive pad **640**. The passivation layer **630** and the conductive pad **640** of the die **600** are respectively similar to the passivation layer **230** and the conductive pad **240** of the die **200**, so the detailed descriptions thereof are omitted herein.

As illustrated in FIG. **2**H, the TSVs **650** are embedded in the semiconductor substrate **610"**. In some embodiments, the TSVs **650** are directly in contact with the conductive patterns **624** to render electrical connection with the interconnection structure **620**. In some embodiments, the bonding vias **670** penetrate through the passivation layer **630** to establish electrical connection with the conductive patterns **624** of the interconnection structure **620**. The bonding layer **660** is formed on the passivation layer **630** and the bonding vias **670**. In some embodiments, the bonding layer **660** includes a dielectric layer **662***a* and a plurality of bonding pads **662***b* embedded in the dielectric layer **662***a*. In some embodiments, the bonding pads **662***b* of the boning layer **660** are electrically connected to the bonding vias **670**. That is, the bonding vias **670** electrically connect the interconnection structure **620** and the bonding pads **662***b* of the bonding layer **660**. The bonding layer **660** and the bonding vias **670** are respectively similar to the bonding layer **280** and the bonding vias **270**, so the detailed descriptions thereof are omitted herein.

In some embodiments, the dies **600** may be capable of performing storage functions. For example, the dies **600** may be Dynamic Random Access Memory (DRAM), Resistive Random Access Memory (RRAM), Static Random Access Memory (SRAM), or the like. However, the disclosure is not limited thereto. In some alternative embodiments, the dies **600** may be Central Process Unit (CPU) dies, Graphic Process Unit (GPU) dies, Field-Programmable Gate Array (FPGA), or the like.

As illustrated in FIG. **2**H, bottom surfaces of the bonding pads **662***b* and a bottom surface of the dielectric layer **662***a* may be collectively referred to as active surfaces AS**2** of the dies **600**. On the other hand, surfaces of the dies **600** opposite to the active surfaces AS**2** may be referred to as rear surfaces RS**2** of the dies **600**. As shown in FIG. **2**H, the bottom surfaces of the bonding pads **662***b* and the bottom surface of the dielectric layer **662***a* are substantially located at the same level height to provide an appropriate active surfaces AS**2** for hybrid bonding.

As illustrated in FIG. **2**H, the dies **600** are individually placed on the corresponding dies **200** such that each die **600** is bonded to the corresponding die **200**. In some embodiments, each die **600** may be bonded to the corresponding die **200** through a hybrid bonding process. In some embodiments, a temperature of the hybrid bonding process ranges from about 150° C. to about 400° C. The hybrid bonding process will be described in detail below.

In some embodiments, the dies **600** may be picked-and-placed onto the active surfaces AS**1** of the dies **200** such that the dies **600** are electrically connected to the dies **200**. In some embodiments, the dies **600** are placed such that the active surfaces AS**2** of the dies **600** are in contact with the active surfaces AS**1** of the dies **200**. Meanwhile, the bonding pads **662***b* of the dies **600** are substantially aligned and in direct contact with the corresponding bonding pads **282***b* of the dies **200**. In some embodiments, to facilitate the hybrid bonding between the dies **600** and the dies **200**, surface preparation for bonding surfaces (i.e. the active surfaces AS**1** and the active surfaces AS**2**) of the dies **600** and the dies **200** may be performed. The surface preparation may include surface cleaning and activation, for example. Surface cleaning may be performed on the active surfaces AS**1**, AS**2** to remove particles on the bonding surface of the dielectric layer **282***a*, the bonding surfaces of the bonding pads **282***b*, the bonding surface of the dielectric layer **662***a*, and the bonding surfaces of the bonding pads **662***b*. In some embodiments, the active surfaces AS**1**, AS**2** may be cleaned by wet cleaning, for example. Not only particles are removed, but also native oxide formed on the bonding surfaces of the bonding pads **282***b* and the bonding pads **662***b* may be removed. The native oxide formed on the bonding surfaces of the bonding pads **282***b* and the bonding pads **662***b* may be removed by chemicals used in wet cleaning processes, for example.

After cleaning the active surfaces AS**1** of the dies **200** and the active surfaces AS**2** of the dies **600**, activation of the bonding surfaces of the dielectric layer **282***a* and the dielectric layer **662***a* may be performed for development of high bonding strength. In some embodiments, plasma activation may be performed to treat the bonding surfaces of the dielectric layer **282***a* and the dielectric layer **662***a*. When the activated bonding surface of the dielectric layer **282***a* is in contact with the activated bonding surface of the dielectric layer **662***a*, the dielectric layer **282***a* of the dies **200** and the dielectric layer **662***a* of the dies **600** are pre-bonded.

After pre-bonding the dies **600** onto the dies **200**, hybrid bonding of the dies **600** and the dies **200** is performed. The hybrid bonding of the dies **600** and the dies **200** may include a thermal treatment for dielectric bonding and a thermal annealing for conductor bonding. In some embodiments, the thermal treatment for dielectric bonding is performed to strengthen the bonding between the dielectric layer **282***a* and the dielectric layer **662***a*. For example, the thermal treatment for dielectric bonding may be performed at temperature ranging from about 200° C. to about 400° C. After performing the thermal treatment for dielectric bonding, the thermal annealing for conductor bonding is performed to facilitate the bonding between the bonding pads **282***b* and the bonding pads **662***b*. For example, the thermal annealing for conductor bonding may be performed at temperature ranging from about 150° C. to about 400° C. After performing the thermal annealing for conductor bonding, the dielectric layer **282***a* is hybrid bonded to the dielectric layer **662***a* and the bonding pads **282***b* are hybrid bonded to the bonding pads **662***b*. For example, the dielectric layer **282***a* is directly in contact with the dielectric layer **662***a*. Similarly, the bonding pads **282***b* are directly in contact with the bonding pads **662***b*. As such, the bonding layer **280** of the dies **200** is hybrid bonded to the bonding layer **660** of the dies **600**. Although FIG. **2**H illustrated that the bonding pads **282***b* and the bonding pads **662***b* have sharp corners (the sidewalls are perpendicular to the top/bottom surfaces), the disclosure is not limited thereto. In some alternative embodiments, after the bonding pads **282***b* are hybrid bonded to the bonding pads **662***b*, corner rounding of the bonding pads may occur. For example, the corners of the bonding pads **282***b* facing the bonding pads **662***b* are rounded. Similarly, the corners of the bonding pads **662***b* facing the bonding pads **282***b* are also rounded. That is, edges of the top surface of each bonding pad **282***b* are rounded. Similarly, edges of the bottom surface of each bonding pad **662***b* are also rounded.

In some embodiments, since the active surfaces AS**1** of the dies **200** are hybrid bonded to the active surfaces AS**2** of the dies **600**, the bonding between the dies **200** and the dies **600** may be considered as face-to-face bonding. In some embodiments, with the presence of the alignment marks **260**, the bonding precision may be effectively improved. That is,

with the aid of the alignment marks **260**, the alignment between the bonding pads **282***b* and the corresponding bonding pads **662***b* may be ensured. It should be noted that although FIG. **2**H illustrated that the sizes (i.e. the widths) of the dies **200** and the dies **600** are substantially identical, the disclosure is not limited thereto. In some alternative embodiments, the size of the dies **200** may differ from the size of the dies **600**. For example, the size of the dies **200** may be larger than the size of the dies **600**. Alternatively, the size of the dies **200** may be smaller than the size of the dies **600**. Moreover, although FIG. **2**H illustrated that the dies **200** and the dies **600** are bonded in a one-to-one manner, the disclosure is not limited thereto. Depending on the size of the die **200**, multiple dies **600** may be bonded to one die **200**.

Referring to FIG. **2**I, an insulating layer **700** is formed over the bonding layer **280** to laterally encapsulate the dies **600**. In some embodiments, a material of the insulating layer **700** includes a molding compound, a polymeric material, such as polyimide, epoxy resin, acrylic resin, phenol resin, BCB, PBO, a combination thereof, or other suitable polymer-based dielectric materials. In some alternative embodiments, the insulating layer **700** may include silicon oxide and/or silicon nitride. In some embodiments, the insulating layer **700** further includes fillers. Alternatively, the insulating layer **700** may be free of fillers. In some embodiments, the insulating layer **700** may be formed by the following steps. First, an insulating material (not shown) is formed over the bonding layer **280** to encapsulate the dies **600**. At this stage, the semiconductor substrates **610"** of the dies **600** are not revealed and are well protected by the insulating material. For example, the rear surface RS**2** of the dies **600** are not revealed. In some embodiments, the insulating material may be formed by a molding process (such as a compression molding process), a spin-coating process, a chemical vapor deposition (CVD) process, a plasma-enhanced chemical vapor deposition (PECVD) process, an atomic layer deposition (ALD) process, or the like. After the insulating material is formed, the insulating material and the semiconductor substrate **610"** are thinned until the TSVs **650** are exposed. In some embodiments, the insulating material and the semiconductor substrates **610"** may be thinned or planarized through a grinding process, such as a mechanical grinding process, a CMP process, or the like. After the thinning process, each die **600** has a thinned semiconductor substrate **610'** and the insulating layer **700** is formed to expose rear surfaces RS**2'** of the dies **600** and the TSVs **650**. That is, rear surfaces RS**2'** of the dies **600** are substantially coplanar with a top surface of the insulating layer **700**. In some embodiments, the TSVs **650** penetrate through the semiconductor substrates **610'**. In some embodiments, the insulating layer **700** may be referred to as "gap fill oxide." After the thinning process, each die **600** has a thickness $H_{600}$ of about 5 μm to about 100 μm. It should be noted that the foregoing process merely serve as an exemplary illustration, and the disclosure is not limited thereto. In some alternative embodiments, the insulating layer **700** may be formed after the semiconductor substrates **610"** are thinned to expose the TSVs **650**.

Referring to FIG. **2**J, a portion of each die **600** is removed to form a plurality of recesses R. For example, a portion of each semiconductor substrate **610'** is removed to form the recesses R. As illustrated in FIG. **2**J, the TSVs **650** are partially located in the recess R. In some embodiments, at least a portion of each TSV **650** protrudes from the semiconductor substrates **610** of the dies **600**. That is, the top surfaces of the TSVs **650** are located at a level height higher than the rear surfaces RS**2"** of the dies **600**. In some embodiments, the semiconductor substrates **610'** may be partially removed through an etching process. The etching process includes, for example, an isotropic etching process and/or an anisotropic etching process. For example, the semiconductor substrates **610'** may be partially removed through a wet etching process, a dry etching process, or a combination thereof.

Referring to FIG. **2**K, a protection layer **680** is formed to fill the recesses R. In some embodiments, the protection layer **680** includes a molding compound, a molding underfill, or the like. Alternatively, the protection layer **680** may be made of a polymeric material, such as polyimide, epoxy resin, acrylic resin, phenol resin, BCB, PBO, or other suitable polymer-based dielectric materials. In some embodiments, the protection layer **680** may include fillers. Alternatively, the protection layer **680** may be free of fillers. As illustrated in FIG. **2**K, the protruding portion of each TSV **650** is laterally encapsulated by the protection layer **680**. The insulating layer **700** surrounds the semiconductor substrate **610**, the interconnection structure **620**, the passivation layer **630**, the bonding layer **660**, and the protection layer **680**. In some embodiments, the protection layer **680** may be considered as part of the dies **600**. In some embodiments, the die **200** and the corresponding die **600** disposed thereon may be collectively referred to as the die stack DS.

As illustrated in FIG. **2**K, a redistribution structure **800** and a plurality of conductive terminals **900** are sequentially formed on the insulating layer **700** and the dies **600** to obtain the reconstructed wafer RW. In some embodiments, the redistribution structure **800** includes a plurality of dielectric layers **810** and a plurality of redistribution conductive layers **820**. The redistribution conductive layers **820** may include a plurality of redistribution conductive patterns. In some embodiments, each redistribution conductive layer **820** is sandwiched between two adjacent dielectric layers **810**. Portions of the redistribution conductive layers **820** may extend vertically within the dielectric layer **810** to establish electrical connection with other overlying or underlying redistribution conductive layers **820**. In some embodiments, a material of the redistribution conductive layers **820** includes aluminum, titanium, copper, nickel, tungsten, combinations thereof, or other suitable conductive materials. For example, the bottommost redistribution conductive layer **820** may include a plurality of copper traces while the topmost redistribution conductive layer **820** may include a plurality of aluminum pads. However, the disclosure is not limited thereto. The redistribution conductive layers **820** may be formed by, for example, electroplating, deposition, and/or photolithography and etching. In some embodiments, a material of the dielectric layer **810** includes polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzooxazole (PBO), or any other suitable polymer-based dielectric material. The dielectric layer **810**, for example, may be formed by suitable fabrication techniques, such as spin-on coating, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), or the like. It should be noted that the number of the dielectric layers **810** and the number of the redistribution conductive layers **820** illustrated in FIG. **2**K are merely for illustrative purposes, and the disclosure is not limited thereto. In some alternative embodiments, the number of the dielectric layers **810** and the number of the redistribution conductive layers **820** may be varied depending on the circuit design. In some embodiments, the redistribution structure **800** is electrically connected to the TSVs **650**. For example, the bottommost redistribution conductive layer **820** physically contact the TSVs **650** to establish electrical

connection with the dies **600**. In some embodiments, the redistribution structure **800** further includes a plurality of under-bump metallurgy (UBM) patterns **830**. The UBM patterns **830** are electrically connected to the redistribution conductive layers **820**. In some embodiments, the UBM patterns **830** are electrically connected to the TSVs **650** through the redistribution conductive layers **820**. In some embodiments, each of the UBM patterns **830** is partially embedded in the topmost dielectric layer **810**.

As illustrated in FIG. **2**K, the conductive terminals **900** are disposed on the UBM patterns **830**. In some embodiments, the conductive terminals **900** are attached to the UBM patterns **830** through a solder flux. In some embodiments, the conductive terminals **900** are, for example, solder balls, ball grid array (BGA) balls, or controlled collapse chip connection (C4) bumps. In some embodiments, the conductive terminals **900** are made of a conductive material with low resistivity, such as Sn, Pb, Ag, Cu, Ni, Bi, or an alloy thereof.

In some embodiments, since known good dies are assembled to the reconstructed wafer RW, the yield of the reconstructed wafer RW may be sufficiently increased at a lower cost. Moreover, by providing die stacks DS in the reconstructed wafer RW, the flexibility in chip size, chip thickness, and chip function integration may be effectively increased. Furthermore, by forming the reconstructed wafer RW with the foregoing process, the pitch of the redistribution conductive patterns in the redistribution structure **800** may be reduced to less than 0.8 μm, thereby achieving fine pitch configuration with larger I/O (input/output) connection.

As mentioned above, the reconstructed wafer RW may undergo further processing to obtain a plurality of packages. FIG. **3** is a schematic cross-sectional view illustrating a package **10** in accordance with some embodiments of the disclosure. Referring to FIG. **3**, a singulation process is performed on the reconstructed wafer RW illustrated in FIG. **2**K to form a plurality of packages **10**. In some embodiments, the dicing process or the singulation process typically involves dicing with a rotating blade or a laser beam. In other words, the dicing or singulation process is, for example, a laser cutting process, a mechanical cutting process, or other suitable processes. In some embodiments, during the singulation process, the dielectric layers **810**, the insulating layer **700**, the dielectric layer **282***a*, the insulating layer **400**, the first sub-layer **252**, the second sub-layer **254**, and the carrier substrate **500** are being cut through. In some embodiments, after the singulation process, the insulating layer **400** may be considered as part of the die **200** while the insulating layer **700** may be considered as part of the die **600**.

As illustrated in FIG. **3**, the die **200** is stacked on the die **600**. In other words, multiple dies **200**, **600** are integrated into a single package **10**. As such, the package **10** may be referred to as a "system on integrated circuit (SOIC) package." In some embodiments, the package **10** may be utilized in other modules/applications, such as chip on wafer on substrate (CoWoS) packaging, flip-chip packaging, integrated fan-out (InFO) packaging, fan-out wafer level packaging (WLP), or the like.

FIG. **4**A to FIG. **4**I are schematic cross-sectional view illustrating a manufacturing process of a reconstructed wafer RW**1** in accordance with some alternative embodiments of the disclosure. Referring to FIG. **4**A, a carrier substrate **110** is provided. A dielectric layer **120**, a dielectric layer **130**, and a bonding layer **284** are sequentially disposed on the carrier substrate **110**. The carrier substrate **110**, the dielectric layer **120**, and the dielectric layer **130** in FIG. **4**A are respectively similar to the carrier substrate **110**, the dielectric layer **120**, and the dielectric layer **130** in FIG. **2**A, so the detailed descriptions thereof are omitted herein.

In some embodiments, the bonding layer **284** includes a dielectric layer **284***a* and a plurality of bonding pads **284***b* embedded in the dielectric layer **284***a*. In some embodiments, a material of the dielectric layer **284***a* includes oxides, such as silicon oxide or the like. Alternatively, the dielectric layer **284***a* may include polyimide, epoxy resin, acrylic resin, phenol resin, benzocyclobutene (BCB), polybenzoxazole (PBO), or any other suitable polymer-based dielectric material. The dielectric layer **284***a*, for example, may be formed by suitable fabrication techniques, such as spin-on coating, CVD, PECVD, or the like. Materials for the bonding pads **284***b* are, for example, aluminum, titanium, copper, nickel, tungsten, or alloys thereof. In some embodiments, a plurality of alignment marks **260** are embedded in the dielectric layer **284***a*. In some embodiments, the alignment marks **260** may be a patterned copper layer or other suitable patterned metal layer. In some embodiments, the alignment marks **260** may be formed by electroplating or deposition. It should be noted that the shapes and numbers of the alignment marks **260** are not limited in the disclosure, and may be designated based on the demand and/or design layout. In some embodiments, the alignment marks **260** are electrically isolated from other components. In other words, the alignment marks **260** are electrically floating.

As illustrated in FIG. **4**A, a plurality of dies **200** is attached to the carrier substrate **110**. In some embodiments, each die **200** includes a semiconductor substrate **210'**, an interconnection structure **220**, a passivation layer **230**, a conductive pad **240**, a plurality of bonding vias **270**, and a bonding layer **282**. The semiconductor substrate **210'**, the interconnection structure **220**, the passivation layer **230**, and the conductive pad **240** in FIG. **4**A are respectively similar to the semiconductor substrate **210'**, the interconnection structure **220**, the passivation layer **230**, and the conductive pad **240** in FIG. **2**A, so the detailed descriptions thereof are omitted herein. On the other hand, the bonding vias **270** and the bonding layer **282** in FIG. **4**A are respectively similar to the bonding vias **270** and the bonding layer **280** in FIG. **2**G, so the detailed descriptions thereof are omitted herein.

In some embodiments, the dies **200** may be capable of performing logic functions. For example, the dies **200** may be Central Process Unit (CPU) dies, Graphic Process Unit (GPU) dies, Field-Programmable Gate Array (FPGA), or the like.

In some embodiments, the dies **200** are picked-and-placed onto the bonding layer **284** such that the bonding layer **282** is adhered to the bonding layer **284** through hybrid bonding. The hybrid bonding process in FIG. **4**A may be similar to the hybrid bonding process shown in FIG. **2**H, so the detailed description thereof is omitted herein. In some embodiments, the dielectric layer **282***a* of the bonding layer **282** is hybrid bonded to the dielectric layer **284***a* of the bonding layer **284**. Meanwhile, the bonding pads **282***b* of the bonding layer **282** are hybrid bonded to and are aligned with the bonding pads **284***b* of the bonding layer **284**. After hybrid bonding the bonding layer **284** to the dies **200**, the alignment marks **260** and the bonding layer **284** may be considered as part of the dies **200**. That is, the bonding layer **284** and the alignment marks **260** are transferred onto the dies **200**. In some embodiments, the bonding layer **282** and the bonding layer **284** may be collectively referred to as a bonding layer **280**. That is, the bonding layer **282** may be referred to as a first sub-layer of the bonding layer **280** while the bonding layer

284 may be referred to as a second sub-layer of the bonding layer 280. In some embodiments, the first sub-layer of the bonding layer 280 is hybrid bonded to the second sub-layer of the bonding layer 280, and the alignment marks 260 are embedded in the second sub-layer.

In some embodiments, the dies 200 are bonded to the carrier substrate 110 in a face down manner. That is, the interconnection structures 200 and the contact pads 240 of the dies 200 face the carrier substrate 110. In some embodiments, the dies 200 are arranged in an array. As illustrated in FIG. 4A, the alignment marks 260 are arranged on a periphery region which surrounds a positioning location of the dies 200. In the other words, the alignment marks 260 are disposed outside of the span of the dies 200. With the presence of the alignment marks 260, the bonding precision may be effectively improved. That is, with the aid of the alignment marks 260, the alignment between the bonding pads 282*b* and the corresponding bonding pads 284*b* may be ensured.

Referring to FIG. 4B, an insulating layer 400 is formed over the bonding layer 284 to laterally encapsulate the dies 200. The insulating layer 400 in FIG. 4B is similar to the insulating layer 400' in FIG. 2B, so the detailed description thereof is omitted herein. During the formation of the insulating layer 400, the semiconductor substrate 210' is thinned or planarized to form the semiconductor substrate 210. The step illustrated in FIG. 4B is similar to the step illustrated in FIG. 2B, so the detailed description thereof is omitted herein. As illustrated in FIG. 4B, the alignment marks 260 are directly in contact with the insulating layer 400. In some embodiments, the insulating layer 400 surrounds the semiconductor substrate 210, the interconnection structure 220, the passivation layer 230, and the bonding layer 282 (the first sub-layer of the bonding layer 280).

Referring to FIG. 4C, a bonding layer 250 is formed on the dies 200 and the insulating layer 400 opposite to the carrier substrate 110. In some embodiments, the bonding layer 250 is a smooth layer having a continuous even surface. In some embodiments, a material of the bonding layer 250 may include silicon oxynitride (SiON), silicon oxide, silicon nitride or the like, and the bonding layer 250 may be formed by deposition or the like. In some embodiments, the bonding layer 250 may be considered as part of the dies 200.

Referring to FIG. 4D, the bonding layer 250 is attached to a carrier substrate 500 opposite to the carrier substrate 110. That is, the dies 200 are bonded to the carrier substrate 500, and the carrier substrates 110, 500 are located on two opposite sides of the dies 200. In some embodiments, the carrier substrate 500 in FIG. 4D is similar to the carrier substrate 500 in FIG. 2D, so the detailed description thereof is omitted herein. In some embodiments, the carrier substrate 500 is adhered to the bonding layer 250 through fusion bonding. The fusion bonding process in FIG. 4C may be similar to the fusion bonding process shown in FIG. 2D, so the detailed description thereof is omitted herein. It should be noted that since the dies 200 are arranged on the carrier substrate 110 in a wafer form and the carrier substrate 500 is in wafer form as well, the bonding between the dies 200 and the carrier substrate 500 may be considered as a wafer-level process. That is, the bonding between the dies 200 and the carrier substrate 500 is a wafer-to-wafer bonding process. In some embodiments, with the presence of the alignment marks 260, the wafer-to-wafer bonding precision may be effectively improved.

Referring to FIG. 4D and FIG. 4E, the carrier substrate 110, the dielectric layer 120, and the dielectric layer 130 are removed from the bonding layer 284. In some embodiments, the carrier substrate 110, the dielectric layer 120, and the dielectric layer 130 are removed through a planarization process, an etching process, a stripping process, the like, or a combination thereof. In some embodiments, bottom surfaces of the bonding pads 284*b* and a bottom surface of the dielectric layer 284*a* may be collectively referred to as active surfaces AS1 of the dies 200. On the other hand, the surface of the bonding layer 250 facing the carrier substrate 500 may be referred to as rear surfaces RS1 of the dies 200. As shown in FIG. 4E, the bottom surfaces of the bonding pads 284*b* and the bottom surface of the dielectric layer 284*a* are substantially located at the same level height to provide an appropriate active surfaces AS1 for hybrid bonding.

Referring to FIG. 4F, a plurality of dies 600 are provided. The dies 600 in FIG. 4F are similar to the dies 600 in FIG. 2H, so the detailed descriptions thereof are omitted herein. In some embodiments, the dies 600 are individually placed on the corresponding dies 200 such that each die 600 is bonded to the corresponding die 200. In some embodiments, each die 600 may be bonded to the corresponding die 200 through a hybrid bonding process. The step illustrated in FIG. 4F is similar to the step illustrated in FIG. 2H, so the detailed description thereof is omitted herein. In some embodiments, the bonding layer 284 (the second sub-layer of the bonding layer 280) is hybrid bonded to the bonding layer 660. That is, the dielectric layer 284*a* of the bonding layer 284 of the die 200 is hybrid bonded to the dielectric layer 662*a* of the bonding layer 660 of the die 600. Meanwhile, the bonding pads 284*b* of the bonding layer 284 are hybrid bonded to and are aligned with the bonding pads 662*b* of the bonding layer 660.

In some embodiments, since the active surfaces AS1 of the dies 200 are hybrid bonded to the active surfaces AS2 of the dies 600, the bonding between the dies 200 and the dies 600 may be considered as face-to-face bonding. In some embodiments, with the presence of the alignment marks 260, the bonding precision may be effectively improved. That is, with the aid of the alignment marks 260, the alignment between the bonding pads 284*b* and the corresponding bonding pads 662*b* may be ensured. It should be noted that although FIG. 4F illustrated that the sizes (i.e. the widths) of the dies 200 and the dies 600 are substantially identical, the disclosure is not limited thereto. In some alternative embodiments, the size of the dies 200 may differ from the size of the dies 600. For example, the size of the dies 200 may be larger than the size of the dies 600. Alternatively, the size of the dies 200 may be smaller than the size of the dies 600. Moreover, although FIG. 4F illustrated that the dies 200 and the dies 600 are bonded in a one-to-one manner, the disclosure is not limited thereto. Depending on the size of the die 200, multiple dies 600 may be bonded to one die 200.

Referring to FIG. 4G to FIG. 4I, by further processing the structure illustrated in FIG. 4F, the reconstructed wafer RW1 is obtained. The steps illustrated in FIG. 4G to FIG. 4I are similar to the steps illustrated in FIG. 2I to FIG. 2K, so the detailed descriptions thereof are omitted herein. In some embodiments, since known good dies are assembled to the reconstructed wafer RW1, the yield of the reconstructed wafer RW1 may be sufficiently increased at a lower cost. Moreover, by providing die stacks DS in the reconstructed wafer RW1, the flexibility in chip size, chip thickness, and chip function integration may be effectively increased. Furthermore, by forming the reconstructed wafer RW1 with the foregoing process, the pitch of the redistribution conductive patterns in the redistribution structure 800 may be reduced

to less than 0.8 μm, thereby achieving fine pitch configuration with larger I/O (input/output) connection.

As mentioned above, the reconstructed wafer RW**1** may undergo further processing to obtain a plurality of packages. FIG. **5** is a schematic cross-sectional view illustrating a package **20** in accordance with some alternative embodiments of the disclosure. Referring to FIG. **5**, a singulation process is performed on the reconstructed wafer RW**1** illustrated in FIG. **4**I to form a plurality of packages **20**. In some embodiments, the dicing process or the singulation process typically involves dicing with a rotating blade or a laser beam. In other words, the dicing or singulation process is, for example, a laser cutting process, a mechanical cutting process, or other suitable processes. In some embodiments, during the singulation process, the dielectric layers **810**, the insulating layer **700**, the dielectric layer **284***a*, the insulating layer **400**, the bonding layer **250**, and the carrier substrate **500** are being cut through. In some embodiments, after the singulation process, the insulating layer **400** may be considered as part of the die **200** while the insulating layer **700** may be considered as part of the die **600**.

As illustrated in FIG. **5**, the die **200** is stacked on the die **600**. In other words, multiple dies **200**, **600** are integrated into a single package **20**. As such, the package **20** may be referred to as a "system on integrated circuit (SOIC) package." In some embodiments, the package **20** may be utilized in other modules/applications, such as chip on wafer on substrate (CoWoS) packaging, flip-chip packaging, integrated fan-out (InFO) packaging, fan-out wafer level packaging (WLP), or the like.

FIG. **6** is a schematic cross-sectional view illustrating a package **30** in accordance with some alternative embodiments of the disclosure. Referring to FIG. **6**, the package **30** in FIG. **6** is similar to the package **10** in FIG. **3**, so similar elements are denoted by the same reference numeral and the detailed descriptions thereof are omitted herein. As illustrated in FIG. **6**, the bonding layer **280** of the die **200** further includes a plurality of conductive patterns **282***c* embedded in the dielectric layer **282***a*. Moreover, the die **600** further includes a plurality of through insulating vias (TSV) **690** penetrating through the insulating layer **700**. In some embodiments, a material of the TSVs **690** includes, aluminum, titanium, copper, nickel, tungsten, and/or alloys thereof. The TSVs **690** may be formed through a plating process. In some embodiments, the TSVs **690** electrically connect the conductive patterns **282***c* of the bonding layer **280** of the die **200** and the redistribution conductive layer **820** of the redistribution structure **800**. In other words, the die **200** may be electrically connected to the redistribution structure **800** through the TSVs **690**.

FIG. **7** is a schematic cross-sectional view illustrating a package **40** in accordance with some alternative embodiments of the disclosure. Referring to FIG. **7**, the package **40** in FIG. **7** is similar to the package **20** in FIG. **5**, so similar elements are denoted by the same reference numeral and the detailed descriptions thereof are omitted herein. As illustrated in FIG. **7**, the bonding layer **284** (the second sub-layer of the bonding layer **280**) of the die **200** further includes a plurality of conductive patterns **284***c* embedded in the dielectric layer **284***a*. Moreover, the die **600** further includes a plurality of through insulating vias (TSV) **690** penetrating through the insulating layer **700**. In some embodiments, a material of the TSVs **690** includes, aluminum, titanium, copper, nickel, tungsten, and/or alloys thereof. The TSVs **690** may be formed through a plating process. In some embodiments, the TSVs **690** electrically connect the conductive patterns **284***c* of the bonding layer **284** of the die **200** and the redistribution conductive layer **820** of the redistribution structure **800**. In other words, the die **200** may be electrically connected to the redistribution structure **800** through the TSVs **690**.

In accordance with some embodiments of the disclosure, a package includes a carrier substrate, a first die, and a second die. The first die includes a first bonding layer, a second bonding layer opposite to the first bonding layer, and an alignment mark embedded in the first bonding layer. The first bonding layer is fusion boded to the carrier substrate. The second die includes a third bonding layer. The third bonding layer is hybrid bonded to the second bonding layer of the first die.

In accordance with some embodiments of the disclosure, a package includes a carrier substrate, a first die, and a second die. The first die includes a first bonding layer and a second bonding layer opposite to the first bonding layer. The first bonding layer is fusion bonded to the carrier substrate. The second bonding layer includes a first sub-layer and a second sub-layer. The first sub-layer is hybrid bonded to the second sub-layer. The second die includes a third bonding layer. The third bonding layer is hybrid bonded to the second sub-layer of the second bonding layer of the first die.

In accordance with some embodiments of the disclosure, a manufacturing method of a reconstructed wafer includes at least the following steps. A first carrier substrate having a first alignment mark formed thereon is provided. A plurality of first dies is attached to the first carrier substrate. The first dies are arranged in an array. The first dies are fusion bonded to the second carrier substrate opposite to the first carrier substrate. The first carrier substrate is removed from the first dies. A first bonding layer is formed onto the first dies. A plurality of second dies are individually placed on the corresponding first dies. Each second die includes a semiconductor substrate, a plurality of through semiconductor vias (TSV) embedded in the semiconductor substrate, and a second bonding layer over the semiconductor substrate. The second bonding layers of the second dies are hybrid bonded to the first bonding layer. The second dies are planarized until the TSVs are exposed.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

**1**. A package, comprising:

a carrier substrate; and

a first die and a second die stacked on the carrier substrate in sequential order, wherein the first die comprises a first bonding layer, a second bonding layer, and an alignment mark embedded in the first bonding layer, the second die comprises a third bonding layer, a surface of the first bonding layer form a rear surface of the first die, a surface of the second bonding layer form an active surface of the first die, the rear surface of the first die is in physical contact with the carrier substrate, and the active surface of the first die is in physical contact with the third bonding layer of the second die.

2. The package of claim 1, wherein the carrier substrate comprises semiconductor materials.

3. The package of claim 1, wherein the second bonding layer and the third bonding layer respectively comprises a dielectric layer and a plurality of bonding pads embedded in the dielectric layer, the dielectric layer of the second bonding layer is in physical contact with the dielectric layer of the third bonding layer, and the bonding pads of the second bonding layer are in physical contact with and are aligned with the corresponding bonding pads of the third bonding layer.

4. The package of claim 3, wherein the first die further comprises:

a semiconductor substrate;

an interconnection structure over the semiconductor substrate;

a passivation layer over the interconnection structure;

a plurality of bonding vias penetrating through the passivation layer and electrically connecting the interconnection structure and the bonding pads of the second bonding layer; and

an insulating layer surrounding the semiconductor substrate, the interconnection structure, and the passivation layer, wherein the second bonding layer is stacked on the passivation layer, the bonding vias, and the insulating layer.

5. The package of claim 4, wherein the alignment mark is in physical contact with the insulating layer.

6. The package of claim 4, wherein the first bonding layer comprises a first sub-layer and a second sub-layer, the first sub-layer is in physical contact with the insulating layer and the semiconductor substrate, a surface of the second sub-layer form the rear surface of the first die, and the alignment mark is embedded in the first sub-layer.

7. The package of claim 3, wherein the second die further comprises:

a semiconductor substrate having a plurality of through semiconductor vias (TSV) embedded therein;

an interconnection structure over the semiconductor substrate and electrically connected to the TSVs;

a passivation layer over the interconnection structure;

a plurality of bonding vias penetrating through the passivation layer and electrically connecting the interconnection structure and the bonding pads of the third bonding layer; and

an insulating layer surrounding the semiconductor substrate, the interconnection structure, the passivation layer, and the third bonding layer.

8. The package of claim 7, wherein the second die further comprises a plurality of through insulating vias (TIVs) penetrating through the insulating layer.

9. A package, comprising:

a carrier substrate; and

a first die and a second die stacked on the carrier substrate in sequential order, wherein the first die comprises a first bonding layer and a second bonding layer, the second die comprises a third bonding layer, a surface of the first bonding layer form a rear surface of the first die, the second bonding layer comprises a first sub-layer and a second sub-layer, a surface of the second sub-layer form an active surface of the first die, the rear surface of the first die is in physical contact with the carrier substrate, and the active surface of the first die is in physical contact with the third bonding layer of the second die.

10. The package of claim 9, wherein the carrier substrate comprises semiconductor materials.

11. The package of claim 9, wherein the first sub-layer, the second sub-layer, and the third bonding layer respectively comprises a dielectric layer and a plurality of bonding pads embedded in the dielectric layer, the dielectric layer of the first sub-layer is in physical contact with the dielectric layer of the second sub-layer, the bonding pads of the first sub-layer are in physical contact with and are aligned with the corresponding bonding pads of the second sub-layer, the dielectric layer of the second sub-layer is in physical contact with the dielectric layer of the third bonding layer, and the bonding pads of the second sub-layer are in physical contact with and are aligned with the corresponding bonding pads of the third bonding layer.

12. The package of claim 11 wherein the first die further comprises:

a semiconductor substrate;

an interconnection structure over the semiconductor substrate;

a passivation layer over the interconnection structure;

a plurality of bonding vias penetrating through the passivation layer and electrically connecting the interconnection structure and the bonding pads of the first sub-layer; and

an insulating layer surrounding the semiconductor substrate, the interconnection structure, the passivation layer, and the first sub-layer.

13. The package of claim 12, wherein the first die further comprises an alignment mark embedded in the second sub-layer, and the alignment mark is in physical contact with the insulating layer.

14. The package of claim 11, wherein the second die further comprises:

a semiconductor substrate having a plurality of through semiconductor vias (TSV) embedded therein;

an interconnection structure over the semiconductor substrate;

a passivation layer over the interconnection structure;

a plurality of bonding vias penetrating through the passivation layer and electrically connecting the interconnection structure and the bonding pads of the third bonding layer; and

an insulating layer surrounding the semiconductor substrate, the interconnection structure, the passivation layer, and the third bonding layer.

15. The package of claim 14, wherein the second die further comprises a plurality of through insulating vias (TIVs) penetrating through the insulating layer.

16. A package, comprising:

a carrier substrate;

a first die disposed on the carrier substrate, comprising:

a semiconductor substrate;

a first bonding layer sandwiched between the semiconductor substrate and the carrier substrate, wherein the first bonding layer extends beyond a span of the semiconductor substrate; and

a first sub-layer and a second sub-layer over the semiconductor substrate, wherein sidewalls of the first sub-layer are aligned with sidewalls of the semiconductor substrate, and the second sub-layer extends beyond a span of the semiconductor substrate; and

a second die stacked on the first die, comprising:

a third bonding layer, wherein the third bonding layer is in physical contact with the second sub-layer.

17. The package of claim 16, wherein the second sub-layer is sandwiched between the first sub-layer and the third bonding layer.

**18**. The package of claim **16**, wherein the first die further comprises an insulating layer laterally surrounding the semiconductor substrate, and the insulating layer is sandwiched between the second sub-layer and the first bonding layer.

**19**. The package of claim **18**, wherein the first die further comprises an alignment mark embedded in the second sub-layer, and the alignment mark is in physical contact with the insulating layer.

**20**. The package of claim **16**, wherein the carrier substrate comprises semiconductor materials.

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